Market Size and the Returns to Surgeon Volume: Evidence from Hip and Knee Replacements

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Abstract

Why are people in larger geographic markets more productive? I investigate one mechanism in the context of hip and knee replacements—the relationship between surgeons' procedural volume and patient health outcomes. Using differential distance as an instrument, I find that surgeon volume affects outcomes and attribute 28% of the benefits of market size to this mechanism. This relationship underscores an externality: patient choice of surgeon affects others' outcomes through surgeon quality. Evaluating a first-best policy and three feasible alternatives, I show the externality increases the welfare effects of these alternatives, but the best achieves only 9% of the first-best gain.

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People in larger geographic markets are more productive. Workers are paid higher wages, innovators produce more patents, and entrepreneurs are more likely to start firms (Glaeser and Maré, 2001; Bettencourt, Lobo, and Strumsky, 2007; Moretti, 2021; Rubinton, 2025). Distinguishing the mechanisms that generate these spatial differences–such as whether they reflect the selection of higher skill individuals into larger markets or some causal effect of being in a large market–is challenging due to endogeneity and data constraints (Glaeser and Gottlieb, 2009).

Understanding these mechanisms is crucial for designing public policy. This knowledge is especially relevant for designing place-based policy in health care, as the federal government spends billions of dollars each year in policies targeted at rural areas (MedPAC, 2022). The optimal policy to improve patient outcomes in small markets depends on the mechanism generating the health benefits of market size. If patients in larger markets have better health outcomes because higher quality physicians choose to practice there, then subsidizing physicians to practice in smaller markets may be optimal. On the other hand, if larger markets improve physician productivity through some agglomerative externality, then policy that leverages these benefits of market size through this externality, such as subsidies for patient transportation, may instead be optimal.

In this paper, I investigate the role of one potential externality–the returns to surgeon volume–in explaining the benefits of market size. Intuitively, if larger markets support more specialization, physicians can perform more of the same procedure, which can improve physician quality and thus generate differences in outcomes between patients in small and large markets. I investigate this idea in the context of hip and knee replacements, two of the most frequently performed medical procedures in the U.S. (Pfuntner, Wier, and Stocks, 2013). Using the differential distance between patients and surgeons as an instrument, I find that surgeon volume significantly affects patient outcomes, and I attribute 28% of the benefits of market size to this returns to surgeon volume mechanism. This result underscores a potentially policy-relevant externality: a patient's choice of surgeon affects other patients' outcomes through the surgeon's quality. Incorporating the externality into a demand model, I evaluate the welfare consequences of a first-best policy and three prominent feasible policies: a minimum volume standard, subsidies for patient transportation, and a policy that moves surgeons to shortage areas.

The health care context, and specifically hip and knee replacement, is an economically meaningful setting to study this question. In 2013, the U.S. spent $24 billion on payments to hospitals for 1.3 million hip and knee replacements, a number that has been increasing in recent years (Molloy et al., 2017; Sloan, Premkumar, and Sheth, 2018). While hip and knee replacements are themselves important to understand, many of the features of this setting, especially the role of surgeon experience, are similar to other surgical procedures (Halm, Lee, and Chassin, 2002). To study this context, I use insurance claims from Medicare, a public health insurance program covering virtually all U.S. individuals sixty-five years and older. Annually, Medicare spends about $12 billion on hip and knee replacements (Molloy et al., 2017).

To begin the analysis, I show three descriptive facts that motivate the returns to surgeon volume as a mechanism underlying the benefits of market size. I first document that patients who

receive care in larger markets have better health outcomes, a pattern that speaks to other settings in which market size is associated with outcomes or productivity (Glaeser and Maré, 2001; Syverson, 2004; Berry and Waldfogel, 2010). Second, I demonstrate that surgeons who choose to practice in larger markets perform more hip and knee replacements. Finally, I show potentially large returns to surgeon volume, as patients who choose surgeons with higher hip and knee volume have better health outcomes (Luft, Bunker, and Enthoven, 1979; Birkmeyer et al., 2002). This last relationship underscores the potential externality of patient choice on other patients that may drive the benefits of market size and influence policy design.

Next, to structure the empirical analysis and evaluate counterfactual policies, I introduce a multinomial logit model of patient demand for surgeons that captures the potential externality implied by the returns to surgeon volume relationship. In the model, patients trade off a surgeon's quality with the distance they must travel for that surgeon, which is the key tradeoff in a setting in which prices do not vary across surgeons. Surgeon quality depends both on an exogenous component, due to innate ability, for example, and an endogenous component through hip and knee volume. Allowing patient utility to depend on a surgeon's hip and knee volume embeds the returns to surgeon volume externality—a patient's choice affects the outcomes of other patients. The model first helps me isolate quasi-random variation in a surgeon's hip and knee volume using differential distance to estimate the returns to surgeon volume. I also use it to understand how patients re-sort to surgeons under alternative policy regimes and thus to compute welfare.

To estimate the returns to surgeon volume, I then use this demand model to construct a control function that relies on differential distance as an instrument. The challenge with estimating the returns to surgeon volume is that patients select surgeons non-randomly, and surgeons select locations non-randomly. Thus, I use the demand model to isolate quasi-random variation in hip and knee volume based on the differential distance between patients and surgeons. Intuitively, some surgeons may have higher hip and knee volume based both on where patients live and other surgeons practice relative to where they practice. Since the first stage is non-linear, I implement a control function approach that generalizes the Heckman selection technique to multiple choices (Dubin and McFadden, 1984; Abdulkadiroğlu et al., 2020; Einav, Finkelstein, and Mahoney, 2022; Mourot, 2024). This empirical strategy assumes that differential distance predicts a patient's choice of surgeon (the first stage) and that differential distance only affects outcomes through the choice of surgeon (the exclusion restriction). I probe the validity of these assumptions and find that the evidence supports them.

I find that surgeon volume significantly impacts patient outcomes and that this relationship explains 28% of the benefits of market size. I first demonstrate that doubling a surgeon's hip and knee volume reduces the probability of a complication by 11%. This result highlights the externality—a patient's choice of surgeon affects other patients' outcomes through the quality of the surgeon. Using the control function approach, I also show that the ordinary least squares (OLS) estimate overstates the true effect in part because patients choose higher exogenous quality surgeons. Hence, re-allocating patients to higher volume surgeons may improve quality, even

without a returns to surgeon volume relationship. Finally, exploring the mechanisms underlying the relationship, I find suggestive evidence that learning-by-doing is the main driver of this effect and that skills depreciate over time (Benkard, 2000; Fitzgerald, Gupta, and Schwab, 2026).

Using the demand model and the estimate of the returns to surgeon volume, I incorporate this externality into a welfare function and estimate the welfare effects of a first-best policy and three prominent feasible policies. Because prices in this setting are administratively set and do not vary across surgeons, this externality is not internalized in prices. Hence, for the first-best policy, I allow a social planner to set surgeon-specific Pigouvian taxes or subsidies to correct for the externality. Meanwhile, for the feasible policies, I first consider a minimum volume standard for surgeons to perform hip and knee replacements, which various states have implemented for stroke and cardiac care. I then consider subsidies for patient transportation, which is increasingly prevalent in the private Medicare Advantage market (Shen et al., 2024). Finally, I consider a policy that moves surgeons to government-designated shortage areas, which mimics other interventions, such as loan forgiveness programs, that financially incentivize physicians to move to shortage areas. Because of the returns to surgeon volume externality, the welfare effects have an ambiguous sign and magnitude. Welfare increases among the patients choosing surgeons who gain volume under these policies but decreases for those choosing surgeons who lose volume.

To provide a relevant comparison for the feasible policies and guide future policy design, I first show that the first-best policy generates large welfare gains as the social planner introduces large subsidies for high-quality surgeons. These subsidies substantially consolidate care. Nevertheless, care consolidates less with the externality than without it. This difference reflects the tradeoff between improving outcomes for patients at highly subsidized surgeons and worsening them for patients at highly taxed surgeons, a tradeoff that is not present without the externality. Finally, I show that the welfare gain from the first-best policy is large, as it exceeds the average amount Medicare pays to a surgeon for hip or knee replacement.

Then, I find that incorporating the returns to surgeon volume externality substantially increases the welfare effects of each of the three feasible policies. Implementing a minimum volume standard mechanically centralizes care, which increases the welfare effect by 31% through the externality. Likewise, the transportation subsidy consolidates care among higher volume surgeons, although the change in concentration is much smaller than the minimum volume standard. Finally, while moving surgeons to shortage areas decentralizes care, welfare actually increases once I incorporate the externality as the surgeons at the movers' origin location experience increases in hip and knee volume. The welfare gain from this volume increase more than offsets the decline in welfare from surgeons who lose volume elsewhere.

I find that the minimum volume standard generates the largest welfare increase among the feasible policies, yet it only achieves 9% of the first-best gain. The minimum volume standard achieves such a large welfare effect relative to the two other feasible policies because it generates large fiscal savings both through the returns to surgeon volume externality and as patients choose higher exogenous quality surgeons. Simply put, the other two policies do not substantially change

patients' choice of surgeon, which is crucial to generate large welfare effects with this externality. Despite these large welfare gains under the minimum volume standard, however, it achieves only a small fraction of the gain under the first-best policy because it is crudely targeted. Lastly, I show that the average effects of some of these policies do mask important heterogeneity between rural and urban patients, but the broad policy implications remain unchanged.

This paper provides key empirical insight into the mechanisms that drive the benefits of agglomeration. Understanding the drivers of these benefits is critical for formulating models of economic geography, such as quantitative urban models, and for evaluating and designing effective public policy, such as local taxation (Greenstone, Hornbeck, and Moretti, 2010; Redding, 2023). Urban economists have long theorized that specialization and learning generate benefits to agglomeration (Duranton and Puga, 2004). In fact, several recent papers strongly suggest that specialization plays a crucial empirical role in driving the benefits of market size, such as in the U.S. labor market or the Italian restaurant industry (Leonardi and Moretti, 2023; Moretti and Yi, 2024). Nevertheless, the endogeneity of location decisions and the lack of microdata on worker tasks have made it difficult to empirically demonstrate that any source, let alone specialization or learning, actually generates benefits to agglomeration (Rosenthal and Strange, 2004; Glaeser and Gottlieb, 2009). This paper contributes to this literature by empirically showing that the returns to surgeon volume–a potential consequence of specialization–are an important mechanism underlying the benefits of agglomeration in health care.

This paper also shows that market size is an important driver of the substantial spatial variation in health outcomes. A wide literature in health economics documents large geographic variation in health outcomes (Chetty et al., 2016; Finkelstein, Gentzkow, and Williams, 2021). While this work documents that supply-side factors are important in explaining a substantial share of this variation, the sources that generate this supply-side variation still remain mostly unknown, despite the importance of understanding these sources to evaluate the welfare consequences of policies to decentralize or centralize the provision of care (Chandra and Staiger, 2007; Deryugina and Molitor, 2021). Perhaps most related to this project, Dingel et al. (2023) studies trade for medical services within the U.S. and finds large scale economies. In exploring the mechanisms underlying the benefits of market size, however, this paper makes an important contribution to understanding what generates these benefits and thus how to design policy.

Finally, this paper contributes causal evidence to the "volume-outcomes" literature and considers what this relationship implies for policy. This literature has documented strong correlations between provider volumes and patient health outcomes, including in the context of orthopedic surgeries (Luft, Bunker, and Enthoven, 1979; Birkmeyer et al., 2002; Halm, Lee, and Chassin, 2002). While a much smaller number of these papers attempts to causally identify this relationship, they either rely on strong identification assumptions, focus on hospitals instead of physicians, or take place in distinctive institutional contexts, such as among deployed surgeons in the military (Gaynor, Seider, and Vogt, 2005; Hentschker and Mennicken, 2018; Avdic, Lundborg, and Vikström, 2019; Lundborg et al., 2021; Fitzgerald, Gupta, and Schwab, 2026). Additionally, these

papers do not consider this "volume-outcomes" relationship as an externality and thus do not incorporate it into a quantitative framework to analyze the policy implications.

1 Context, data, and measurement

I investigate the role of the returns to surgeon volume in explaining the benefits of market size in the context of hip and knee replacements, two of the most commonly performed medical procedures in the U.S. Since many patients who undergo a hip or knee replacement are elderly, this context lends itself well to the dataset I use, insurance claims from the Medicare program.

1.1 Context: Total hip and knee replacement

Hip and knee replacements are frequent medical procedures and comprise a large share of U.S. health care costs. Each year, surgeons perform about 1.3 million of these replacements. In 2010, 2.3% of the U.S. population was living with a hip or knee replacement, and this number jumps to 13.7% among those 70 years or older (Kremers et al., 2015). The number of hip and knee replacements is also growing over time. From 2000 to 2014, the annual number of hip and knee replacements per 100,000 people increased by over 100% (Sloan, Premkumar, and Sheth, 2018). Insurers pay hospitals alone about $24 billion each year for care, comprising 0.8% of health care expenditures (Molloy et al., 2017). In the Medicare context, roughly 7% of discharges and 8.5% of inpatient spending, or $12 billion annually, are for hip and knee replacements. In addition, while complication rates from hip and knee replacement are low, the costs of complications are quite large. For instance, 7% of complications result in death, and Medicare pays twice as much for a hip or knee replacement with an infection as for one without (Yi et al., 2015).

Total hip and knee replacement, or more formally total hip and knee arthroplasty, are surgical procedures in which an orthopedic surgeon removes a patient's hip or knee joint and replaces it with an artificial implant. Patients who have severe joint arthritis and who do not respond well to more conservative treatments often choose to undergo replacements. Thus, they are almost always elective. The procedures are typically successful at eliminating pain and improving patient mobility so patients can enjoy a higher quality of life (Räsänen et al., 2007). Orthopedic surgeons generally perform these replacements, and hip and knee replacements are two of the most common surgeries they perform. Orthopedic surgeons are specialized surgeons, who have long training periods and earn higher income compared to physicians in other specialties (Gottlieb et al., 2023). Many orthopedic surgeons receive fellowship training or subspecialize in both hip and knee replacements in part because of their procedural similarity, which is why I consider these two procedures together.

The typical patient chooses an orthopedic surgeon through referrals and word-of-mouth reviews. Primary care physicians or rheumatologists often refer their severely arthritic patients to orthopedic surgeons for more intensive treatments. At the same time, the typical patient learns about orthopedic surgeons through word-of-mouth reviews, such as from family members, friends, or even people at the same church. Because hip and knee replacements are elective and there are virtually no network restrictions in this context, patients can in theory travel quite far for a sur-

geon. However, the median patient only travels ten miles for a replacement. Additionally, because prices do not vary across surgeons in this context, patients do not consider prices in their choice.

Once a patient chooses a surgeon, the surgeon's skill and experience play an important role in determining patients' outcomes. First, the surgeon orders physical exams, X-rays, and lab tests for their patient, which the surgeon uses to prepare for any potential surgical risks. Then, while operating, the surgeon's most salient concern is an infection, which is more likely to occur the longer the wound is open. As surgeons gain skill and experience, they can operate more quickly, reducing the probability of adverse health consequences (Lundborg et al., 2021; Nairn et al., 2021). After surgery, a patient typically remains in the hospital for one to three days and then begins months of physical therapy.[1] Throughout this process, other providers, including anesthesiologists, nurses, and physical therapists, also assist with care. Thus, their skill and the team-specific human capital between them and the surgeon may also affect the patient's outcomes (Chen, 2021). While hip and knee replacements are themselves important to understand, many of the features of this context, especially the role of surgeon experience, are similar to other surgical procedures, such as cardiovascular surgeries (Halm, Lee, and Chassin, 2002).

1.2 Data and sample construction

For my primary dataset, I use a 20% random sample of insurance claims from the Medicare program. The dataset consists of claims, or billing statements, that providers submit to Medicare Fee-For-Service (FFS), the traditional Medicare coverage that is publicly provided.[2] Specifically, I use professional claims typically submitted by physicians, outpatient claims, and institutional hospital and nursing home stays. The data includes patient, physician, and hospital identifiers and procedural and diagnostic medical codes. I link these claims to other data from the Centers for Medicare & Medicaid Services (CMS) that gives basic demographic information, such as age and sex, for both patients and surgeons. The dataset spans the years starting in 2006 and ending in 2019.

I use the Medicare FFS data primarily because it includes a large share of hip and knee replacements, physicians, and geographies. Specifically, Medicare FFS covers about 40% of all hip and knee replacements and includes essentially all orthopedic surgeons in the U.S. Finally, this dataset provides nationwide geographic coverage of patients and surgeons. It has such impressive coverage because there are minimal network or geographic restrictions for participation in the Medicare FFS program (both for surgeons and patients). The dataset's key limitations are that it lacks detailed clinical information and does not include non-Medicare FFS claims, which are two challenges that I overcome with my empirical strategy.

To complete the dataset, I supplement the Medicare claims with other data. First, I augment

[1] Many of these procedures now take place in the outpatient setting. For all but two years of the sample period, these procedures were on the "inpatient-only list," implying that Medicare would not reimburse providers if these replacements were billed in an outpatient setting. In 2018, knee replacements were moved off the inpatient-only list, and hip replacements followed in 2020.

[2] While the private version of Medicare, Medicare Advantage, has been rapidly growing in the past fifteen years, FFS still covers 66% of Medicare patients in the last year of my data.

the claims data with annual, publicly available Provider of Service (POS) files, which give the ZIP code for all Medicare-certified hospitals, and thus the place of surgery. For various robustness exercises, I incorporate publicly available data on physicians' medical school from the CMS National Downloadable File, which I link to U.S. News and World Report medical school rankings in 2018. For the same purpose, I also collect data on Census block group amenities using the Environmental Protection Agency (EPA) Smart Location Database. Meanwhile, for the counterfactual policy analysis, I employ county-level wage data from the Bureau of Labor Statistics (BLS) Quarterly Census of Employment and Wages (QCEW) and data on Health Professional Shortage Area (HPSA) designations. Finally, I obtain inpatient, outpatient, and emergency department data files for all patients in the state of Florida in 2018. While I cannot link this dataset to my primary sample, I use it to better understand the relationship between surgeons' Medicare FFS hip and knee volume and their all-payer volume.

To adapt the Medicare FFS data to my setting, I define my sample and make two primary sample restrictions. The initial sample includes all hip or knee replacements performed by orthopedic surgeons from 2006–2019.[3] For the first sample restriction, I require surgeons to perform at least ten procedures in my initial sample (97.6% of hip and knee replacements) to avoid measurement error from improperly identifying the performing surgeon and to meaningfully compute quality metrics. Second, to simplify the empirical analysis, I restrict to patients who receive a hip or knee replacement for the first time (79.6% of the remaining hip and knee replacements).

1.3 Measurement and summary statistics

I measure volume as a surgeon's number of Medicare FFS hip and knee replacements in the 365 days prior to a patient's surgery date and consider this measure as a proxy for a surgeon's recent experience. I define volume in this way for two main reasons. First, as shown in Appendix Figure A1, correlating lagged measures of volume with patient health outcomes suggests that recent volume is more important than distant volume for patient outcomes. In fact, the relationship between volume and outcomes dies out after just one lag. Second, this definition follows the "volume-outcomes" literature, in large part because the limited time frame of datasets prohibits the use of cumulative volume (Birkmeyer et al., 2002; Halm, Lee, and Chassin, 2002). Despite this limitation, annual Medicare FFS hip and knee volume proxies well for cumulative Medicare FFS hip and knee volume.[4] Additionally, using the external Florida data, I show that a surgeon's Medicare FFS hip and knee volume is a strong proxy for all-payer hip and knee volume.[5]

To measure patient health outcomes, I rely on a previously established measure of surgery-related complications. I directly employ this measure from the Yale New Haven Health Services

[3] I exclude partial replacements and revisions from the sample, as they are procedurally distinct from and more complex than total hip or knee replacements.

[4] Among the surgeons for whom I can observe cumulative Medicare FFS hip and knee volume, the correlation between this measure and annual Medicare FFS hip and knee volume is 0.77.

[5] The cross-sectional correlation between a surgeon's Medicare FFS hip and knee volume and total hip and knee volume is 0.95. Also, regressing log total hip and knee volume on log Medicare FFS hip and knee volume yields a coefficient of 1.06.

Corporation – Center for Outcomes Research and Evaluation, which developed it for CMS. Specifically, this measure is a dichotomous indicator if a patient experiences one of eight surgery-related complications, such as an infection, within a specific time span following the replacement (CMS, 2022). While complications are infrequent, they are extremely serious adverse outcomes, as death occurs in 7% of complications. The three most common complications are periprosthetic joint infection, wound infection, or other wound complication (34.2%); pneumonia or other respiratory complications (18.9%); and pulmonary embolism (15.5%). All eight complication types and their frequency are listed in Appendix Table A1. Following CMS' methodology, I adjust this complication measure for observable patient covariates, such as age, comorbid diseases, and indicators of patient frailty that are clinically relevant and related to the outcome. I list all of these covariates in Appendix Table A2. As others have shown, complications exhibit a strong negative correlation with self-reported quality of life, which is a key metric used to assess the surgery's success and a criterion on which patients select surgeons (Askari et al., 2024).

Finally, for geographic measurement, I employ two commonly used health care geographic units and formally define market size and distance. For geographic markets, I employ Hospital Referral Regions (HRRs) and Hospital Service Areas (HSAs), which are based on Medicare FFS patients' travel for care. HRRs are regional health care markets for tertiary medical care, whereas HSAs are local health care markets for hospital care. In my sample, 83% of patients receive a hip or knee replacement within their HRR, while 55% receive one within their HSA, which is consistent with other health care settings (Dingel et al., 2023). Among markets with any hip and knee replacement in my sample, the median HRR contains thirteen hospitals in which a surgeon performs a hip and knee replacement, while the median HSA contains one such hospital. I define market size as the number of Medicare FFS patients within an HSA, and I measure distance as the as-the-crow-flies distance between ZIP code centroids.

To provide a better picture of the patients and orthopedic surgeons in the sample, I discuss descriptive statistics for the patients and orthopedic surgeons. Examining these statistics for patients in Table 1 shows that the typical patient is seventy-two years old, female, and white. 7% of patients are dually eligible for Medicaid, a public health insurance program primarily for low-income individuals and thus a proxy for low income, and the median patient travels ten miles for a replacement. Additionally, complications occur in 3.1% of replacements. Meanwhile, there are 11,643 surgeons in the sample of which 98% are male. The median surgeon performs twenty-six hip and knee replacements annually, but surgeon hip and knee volume is heavily right-skewed, as shown in Appendix Figure B1. Using the external Florida data, I calculate that the average physician's Medicare FFS share of total hip and knee replacements is 37% in 2018.

	Mean or median
Age	72.4
Fraction male	0.36
Fraction white	0.91
Fraction dual eligible	0.07
Fraction knee	0.66
Complication rate	0.031
Distance (median)	10.4
N	711,341

Table 1: Patient descriptive statistics

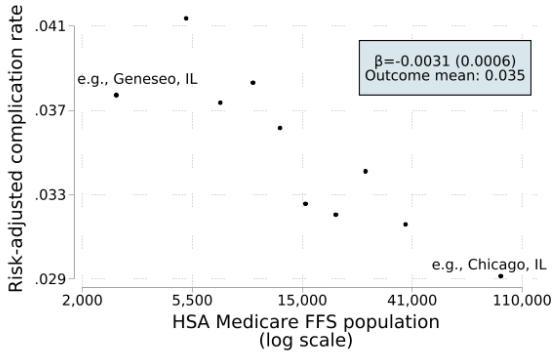
Notes: This table shows descriptive statistics for all patients in my sample. All statistics are averages except for distance, which is the median. Distance is measured in miles.

2 Descriptive evidence: The returns to surgeon volume as a mechanism

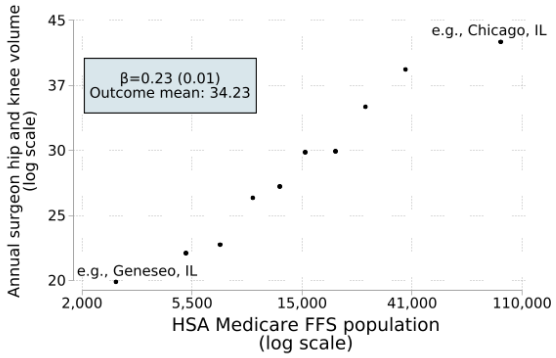
To motivate the returns to surgeon volume as a mechanism underlying the benefits of market size, I first demonstrate that patients who receive care in larger markets have better health outcomes. Figure 1a shows that doubling the size of a market corresponds to a 9% decline in the risk-adjusted complication rate. For example, in the first decile of market size in a market like Geneseo, Illinois, the probability of a complication is 3.8%. Nevertheless, it falls to 2.9% in the last decile for large cities like Chicago, which is a substantial drop given the large cost of a complication. In Appendix Table B1, I demonstrate that including patient risk covariates actually increases the size of this relationship and that differences in patients across HRRs, selection into traveling outside the market, and different behavior after discharge do not appear to drive this relationship.[6] While other work has found similar correlations for mortality or life expectancy, the magnitude and shape of this relationship are mostly unknown (Finkelstein, Gentzkow, and Williams, 2021; Deryugina and Molitor, 2021). This descriptive pattern also speaks to other settings in which market size appears to drive outcomes or productivity, such as wages or restaurant quality (Glaeser and Maré, 2001; Syverson, 2004; Berry and Waldfogel, 2010).

I then show that the average surgeon in a larger market performs more hip and knee replacements than the average surgeon in a smaller market. Namely, Figure 1b demonstrates that doubling the size of a market is associated with a 23% increase in the average surgeon's hip and knee volume. Using the same example as before, surgeons perform on average twenty-five hip and knee replacements annually in the first decile of market size, such as in Geneseo, as compared to forty-six in the last decile, such as in Chicago. This descriptive pattern relates to other work that has documented that physicians specialize more in larger markets due to a greater division of labor (Baumgardner, 1988; Dingel et al., 2023). This phenomenon even extends to other occupations and industries, such as lawyers, and is considered a potentially important microfoundation un-

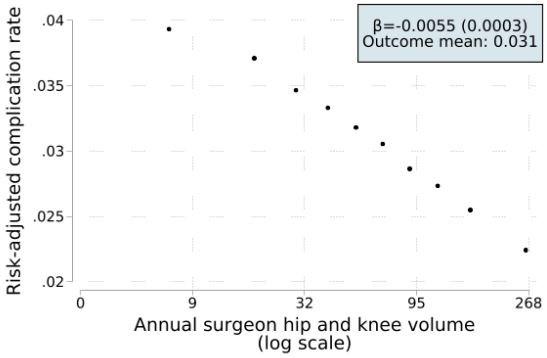
[6] This result is consistent with Chandra and Staiger (2007), who argue that in areas that specialize more in a given procedure (in this context, the large markets), the patients who select into treatment will be less clinically appropriate for such a procedure and thus have worse health outcomes.



(a) Panel A: Relationship between complication rate and market size



(b) Panel B: Relationship between surgeon hip and knee volume and market size



(c) Panel C: Relationship between complication rate and surgeon hip and knee volume

Figure 1: Motivating the returns to surgeon volume as a mechanism underlying the benefits of market size

Notes: Figure 1a shows the relationship between the probability of a risk-adjusted complication and market size within an HSA. The probability of a risk-adjusted complication for an HSA is the HSA fixed effect from a linear probability model in which a binary indicator for a complication is regressed on all the patient risk covariates in Appendix Table A2 and HSA fixed effects for where the patient receives care. For the same HSAs, Figure 1b shows the relationship between a surgeon's average hip and knee volume and the market size of their primary practice HSA. Surgeons are assigned to HSAs based on where they perform the plurality of hip and knee replacements over the whole sample period. Market size is defined as the HSA's average annual market size over the sample period. In Panels 1a and 1b, each HSA is binned into ten equal-sized bins based on market size. Panel 1c shows the relationship between a binary indicator for a complication and a surgeon's annual hip and knee volume, controlling for the patient covariates in Appendix Table A2. Each hip and knee replacement is binned into 10 equal-sized bins based on the surgeon's hip and knee volume.

derlying the benefits of agglomeration (Duranton and Puga, 2004; Garicano and Hubbard, 2007).

Finally, I show a suggestive returns to surgeon volume relationship, as patients who choose higher hip and knee volume surgeons have better outcomes. Figure 1c shows that doubling a surgeon's hip and knee volume corresponds to an 18% reduction in the probability of a risk-adjusted complication. For example, when treated by surgeons in the first decile of hip and knee volume, patients have a complication in 3.9% of replacements, as compared to 2.2% when treated by surgeons in the top decile of hip and knee volume.[7] This relationship echoes the results from the large "volume-outcomes" literature (Luft, Bunker, and Enthoven, 1979; Birkmeyer et al., 2002). It also touches on a broader literature that emphasizes the important role of learning and experience for firm and worker productivity (Arrow, 1962; Irwin and Klenow, 1994; Levitt, List, and Syverson, 2013; Dinerstein, Megalokonomou, and Yannelis, 2022). This relationship underscores a policy-relevant externality—patient choice of surgeon affects other patients' outcomes. Finally, this figure demonstrates that complications are roughly linear in log volume, which affirms the intuitive notion that performing one more replacement has a larger effect on patient outcomes when a surgeon's volume is low and motivates subsequent functional form choices.

Taking the descriptive returns to surgeon volume as causal, these three figures together indicate that the returns to surgeon volume explain 46% of the benefits of market size. Namely, holding fixed the spatial distribution of surgeons, the relationships from Figures 1b and 1c imply that doubling market size is associated with a 4.1% decline in the probability of a complication. Thus, using the estimate from Figure 1a, the returns to surgeon volume explain 46% of the benefits of market size, suggesting that policy design may benefit from incorporating the returns to surgeon volume externality. Such a conclusion, however, is premature given that the returns to surgeon volume estimated here is not necessarily causal. In fact, Appendix Table B2 strongly suggests that it is biased upward, as adding patient covariates and hospital or surgeon fixed effects all decrease the magnitude of the relationship. Nevertheless, 46% may also be a lower bound, if, for instance, surgeons with higher hip and knee volume operate on unobservably riskier patients. In the subsequent section, therefore, I describe the model and empirical strategy I use to obtain the causal effect of surgeon volume on patient outcomes.

3 Model and empirical strategy

To both causally identify the returns to surgeon volume and assess the welfare consequences of various counterfactual policies, I introduce a model of patient demand for surgeons that allows for the returns to surgeon volume externality. I then introduce a model of the returns to surgeon volume and show how the demand model allows me to identify the returns to surgeon volume using differential distance as an instrument.

[7] In Appendix Table B3, I show similar relationships when using the thirty-day readmission rate or the length of stay as health outcomes.

3.1 Model of patient demand for surgeons

I introduce a multinomial logit model of patient demand for surgeons in which patients trade off a surgeon's quality, which depends on other patients' choices through volume, with travel distance. I specify patient i's utility from surgeon j as follows:

$$u_{ij} = \delta_j(v_{ij}) - \tau \ln d_{ij} + \eta_{ij}, \tag{1}$$

where $\delta_j(v_{ij})$ is the average patient utility from surgeon j given surgeon j's hip and knee volume v_{ij} in the 365 days prior to patient i's surgery date, d_{ij} is the distance (plus one) between patient i and surgeon j, and η_{ij} is an idiosyncratic demand shock drawn *i.i.d.* from a type-1 extreme value distribution. Intuitively, distance here acts as the price, since Medicare FFS patients pay the same financial cost regardless of the surgeon they choose. Taking the log of distance reflects the fact that patients bear less of a utility cost for one additional mile at larger distances, a common assumption in the health care literature that I show empirical support for in Appendix Figure C1 (Einav, Finkelstein, and Mahoney, 2022). The logit assumption on the error term aids the empirical strategy and allows for an easier evaluation of the policy counterfactuals. This utility function abstracts from the referral process, so patient i's choice can be interpreted as a joint decision between the patient and the patient's care team, such as a primary care physician.

To incorporate the returns to surgeon volume externality, I allow the average utility from a surgeon to depend on both exogenous quality and endogenous quality through volume. Specifically, I estimate demand using log hip and knee volume as a linear shifter of $\delta_j(v_{ij})$, such that:

$$\delta_j(v_{ij}) = \delta_{0j} + \delta_1 \ln v_{ij}, \tag{2}$$

where v_{ij} is the hip and knee volume (plus one) of surgeon j in the 365 days prior to patient i's surgery date.[8] The parameter δ_{0j} captures the demand for surgeon j's exogenous quality, due to innate ability, for instance. Meanwhile, the parameter δ_1 reflects the returns to surgeon volume relationship, as it captures demand for endogenous quality due to surgeon j's hip and knee volume.[9] I express $\delta_j(v_{ij})$ as a function of log volume to reflect diminishing utility gains from one additional procedure, which is empirically reflected in Figure 1c. Demand for exogenous surgeon quality, δ_{0j}, is identified based on how far patients travel for surgeon j, relative to other surgeons in the market. Demand for a surgeon's hip and knee volume, δ_1, is identified based on travel to surgeon j when the surgeon has different volume—if patients travel farther to the surgeon when the surgeon has higher hip and knee volume, then δ_1 is positive.[10] Due to potential latent choice

[8] I add one to volume because some surgeons have not performed any hip or knee replacements in the previous 365 days. In Appendix Tables D1 and D2, I show robustness of the relevant demand model estimates to not adding one.

[9] δ_1 also captures demand for other characteristics correlated with volume. I return to the interpretation later.

[10] The surgeon fixed effects may exacerbate downward bias in δ_1 due to measurement error (Griliches and Hausman, 1986). If anything, this bias would attribute more of the variation in demand for surgeons to their exogenous quality, rather than their volume. Thus, it would understate the returns to surgeon volume relationship and the welfare effects of the counterfactual policies.

constraints, such as available capacity or information asymmetries, I interpret the demand parameters as capturing both demand and these constraints (Agarwal and Somaini, 2022).[11]

3.2 Empirical strategy to identify the returns to surgeon volume

To estimate the returns to surgeon volume, I first relate patients' health outcomes to their surgeon's hip and knee volume with various controls. For patient i and surgeon j, I consider the following linear probability model:

$$\begin{aligned} \text{comp}_{ij} &= \beta_0 + \beta_1 \ln v_{ij} + X_i\alpha + \gamma_{t(i)} + \epsilon_{ij} \\ \epsilon_{ij} &= \xi_j + \psi_i + \lambda_{ij} + \omega_{ij} \end{aligned} \tag{3}$$

where comp_{ij} equals one if patient i has a complication, v_{ij} is surgeon j's hip and knee volume (plus one) in the 365 days prior to patient i's surgery date, X_i are the patient risk covariates listed in Appendix Table A2, and $\gamma_{t(i)}$ are fixed effects for the calendar year in which patient i receives the surgery. I also include four different unobserved error terms: surgeon-specific exogenous quality ξ_j, unobserved patient characteristics ψ_i, patient-surgeon match-specific characteristics λ_{ij}, and an idiosyncratic error ω_{ij}. I use log volume for the functional form because of Figure 1c, although I show robustness to other functional forms.[12] β_1 is the coefficient of interest in this specification, yet the three non-idiosyncratic error terms, ξ_j, ψ_i, and λ_{ij}, threaten causal identification.

Prior to discussing the empirical strategy, I highlight that the true returns to surgeon volume parameter, β_1, captures the bundled treatment effect from a patient choosing a higher volume surgeon. That is, the causal β_1 captures not just the "pure" effect of a surgeon's hip and knee volume on outcomes but other mechanisms that potentially accompany changes in volume, such as higher team-specific human capital with nurses, additional human capital investments, or congestion costs. In this setting, this effect is the policy-relevant parameter of interest because I consider policy that funnels patients to a specific surgeon, which may also coincide with changes in team-specific human capital, for instance. In the results section, however, I investigate some of these mechanisms underlying the returns to surgeon volume relationship.

To build intuition for the empirical strategy, I present examples of the types of selection that threaten causal identification for the three non-idiosyncratic error terms. First, if patients select surgeons with higher exogenous quality or higher exogenous quality surgeons practice in larger markets, β_1 would be biased upward due to selection on ξ_j. While surgeon fixed effects would address this selection, insufficient variation in hip and knee volume introduces attenuation bias

[11] For the empirical strategy, this interpretation does not present an issue, as I seek to recover the parameters that capture the characteristics on which patients actually select surgeons. I thus address these latent choice constraints explicitly in the counterfactual policy analysis.

[12] I also estimate Equation 3 using a constant relative risk aversion (CRRA) parameterization, such that I replace $\ln v_{ij}$ with $\frac{v_{ij}^{1-\zeta}-1}{1-\zeta}$. Recall the CRRA specification nests the log specification, as it approaches $\ln v_{ij}$ when $\zeta = 1$. I estimate ζ using maximum likelihood estimation and find that $\zeta = 0.80$ with a 95% confidence interval of $[0.68, 0.92]$. While the log specification is statistically rejected, this exercise still indicates that the log specification is a reasonable approximation in this setting.

when volume is mismeasured (Griliches and Hausman, 1986).[13] Additionally, since annual hip and knee volume is a proxy for a surgeon's experience, then a within-surgeon estimate is challenging to interpret. Second, patients may select surgeons based on unobservable patient characteristics, as captured in ψ_i, such as their underlying risk. For example, if unobservably less risky patients select surgeons with higher hip and knee volume, β_1 would be biased upward. Finally, patients may select surgeons based on an idiosyncratic match (Roy selection), as captured in λ_{ij}. For example, if higher hip and knee volume surgeons specialize in treating patients with heart conditions and patients with heart conditions choose those surgeons, then β_1 would be biased upward.

To address this non-random selection, I construct a control function that relies on the differential distance between patients and surgeons as an instrument. Intuitively, some surgeons may have higher hip and knee volume based both on where patients live and other surgeons practice relative to where they practice. To implement this approach, I use the demand model as a first stage to construct a control function in which the vector of distances between a patient and each surgeon functions as the instrument. Following Dubin and McFadden (1984), this approach generalizes Heckman selection to multinomial selection models with many alternatives (Heckman, 1976).

Formally, I construct a control function with three types of controls that explicitly address the three non-idiosyncratic error terms that may bias the returns to surgeon volume. That is, I impose the following structure on the conditional expectation of the estimating equation error term, ϵ_{ij}, from Equation 3:

$$\mathbb{E}\left[\epsilon_{ij}|v_{ij}, X_i, \gamma_{t(i)}, \widehat{\delta}_{0j}, \eta_{i1}, \cdots, \eta_{iJ}, D_i = j\right] = \kappa\widehat{\delta}_{0j} + \sum_k \phi_k \tilde{\eta}_{ik} + \varphi\tilde{\eta}_{ij}, \tag{4}$$

where $\widehat{\delta}_{0j}$ is the estimated demand for exogenous surgeon quality from the demand model in Equation 1, $\tilde{\eta}_{ij} = \eta_{ij} - \mu_\eta$ are the mean-zero logit shocks from the demand model, J is the total number of surgeons, and D_i denotes patient i's chosen surgeon. This functional form assumption closely follows the recent literature building on the Dubin and McFadden (1984) approach and can be thought of as a first-order approximation of a specification that allows for a richer relationship between choice and outcomes (Abdulkadiroğlu et al., 2020; Einav, Finkelstein, and Mahoney, 2022; Mourot, 2024).

Intuitively, these three types of controls allow for correlations between patients' health outcomes and both observed surgeon quality and the unobserved patient or patient-surgeon characteristics on which patients may select surgeons. The first control, $\widehat{\delta}_{0j}$, addresses selection on exogenous surgeon quality, ξ_j. The basic idea is that it controls for the relative distance that patients travel for the surgeon, independent of the travel for the surgeon's volume, which provides infor-

[13] In my sample, within-surgeon variation in hip and knee volume explains only 22% of the total variation in volume. Measurement error in hip and knee volume due to the fact that I only observe a 20% sample of Medicare FFS hip and knee volume only exacerbates this attenuation bias. This issue is reflected in the "volume-outcomes" literature, as many papers omit provider fixed effects (Lundborg et al., 2021).

mation on the surgeon's exogenous quality. Meanwhile, the last two types of controls, the $\tilde{\eta}_{ik}$'s and $\tilde{\eta}_{ij}$, are the residuals from the first stage. The $\tilde{\eta}_{ik}$'s address selection on unobserved patient factors, ψ_i. Intuitively, a patient's unobserved preference for every surgeon, as revealed through relative travel distance, provides information on this unobservable risk. Likewise, $\tilde{\eta}_{ij}$ addresses selection on match characteristics, λ_{ij}, as a patient's unobserved preference for the chosen surgeon, also as revealed through relative travel distance, provides information on this match.

Since the error terms are unobservable, I use the distributional assumptions to obtain closed-form selection correction controls and show that the vector of distances between a patient and each surgeon serves as the instrument. Namely, since I assume the error terms are drawn *i.i.d.* from a logit distribution, I can compute their expected values conditional on the choice, yielding the following conditional expectation function:

$$\mathbb{E}\left[\epsilon_{ij}|v_{ij}, X_i, \gamma_{t(i)}, \widehat{\delta}_{0j}, d_i, D_i = j\right] = \kappa\widehat{\delta}_{0j} + \sum_k \phi_k \theta_{ik}\left(j, d_i\right) + \varphi\theta_{ij}\left(j, d_i\right), \tag{5}$$

where $d_i = (d_{i1}, \cdots, d_{iJ})$ is the vector of distances between patient i and each surgeon that functions as the instrument.[14] Meanwhile, $\theta_{ik}(j, d_i)$ are functions of the logit choice probabilities from the demand model:

$$\theta_{ik}\left(j, d_i\right) = \begin{cases} -\ln \widehat{p}_{ik}\left(d_i\right), & k = j \\ \frac{\widehat{p}_{ik}(d_i)}{1-\widehat{p}_{ik}(d_i)} \ln \widehat{p}_{ik}\left(d_i\right), & \text{otherwise} \end{cases}, \tag{6}$$

where $\widehat{p}_{ik}$ is the predicted probability that patient i chooses surgeon k. I show these derivations in Appendix C.2. Note that the conditional expectation function in Equation 5 now conditions on the (excluded) instrument, the distance between a patient and every surgeon. Provided differential distance is a valid instrument, these selection correction terms allow me to recover the causal returns to surgeon volume.

To provide more intuition for the approach, I discuss comparisons between this control function and a two-stage least squares (2SLS) implementation. Similar to a 2SLS implementation, the relevance assumption and exclusion restriction must hold. As opposed to the 2SLS setting, however, I microfound distance as an instrument using the demand model, rather than instrumenting for hip and knee volume with a more arbitrary function of distance, which implicitly imposes strong functional form assumptions. The demand model, in turn, allows me to address selection on exogenous surgeon quality, which the volume-outcomes literature often does not address. Additionally, I obtain more precise estimates with the control function approach (Wooldridge, 2015). Finally, this approach allows me to understand the types of selection that bias the OLS estimate. These advantages, though, come at the cost of a greater reliance on the functional form assumptions, for which I show support when discussing identification in Section 3.4. Unlike the 2SLS implementation, this approach also does not correct for measurement error in volume, so I em-

[14] Note that I take the expectation of the true η's because the expectation of the predicted η's are by definition mean-zero. Intuitively, the mean of the predicted η's just captures noise, not selection on unobservable characteristics.

ploy an alternative measurement error correction strategy, which I describe in Appendix C.4.

3.3 Estimation

To ease the computational burden of estimating the demand model, I restrict patients' choice sets to surgeons within their geographic market. Namely, I define a patient's choice set as all surgeons who perform a hip or knee replacement within the patient's HRR of residence in the 365 days before or after the patient's surgery date. I assign surgeons to the ZIP code based on where and when they perform most of their hip and knee replacements within this time frame.[15] I define the outside option as choosing a surgeon outside of a patient's HRR, which occurs in 17% of all hip and knee replacements. For the outside option, I normalize utility to be zero. The median patient has forty-four surgeons in the choice set, and the median surgeon is twenty-nine miles away.

Since the demand model is now estimated within a market, I modify the estimating equation slightly by adding fixed effects for a surgeon's practice location. Specifically, the δ_{0j}'s are now only relative to other surgeons within the same geographic market because of the market-level demand estimation. Thus, denoting a surgeon's HRR as m, the main estimating equation becomes:

$$\text{comp}_{ijm} = \beta_0 + \beta_1 \ln v_{ij} + X_i\alpha + \gamma_{t(i)} + \kappa\widehat{\delta}_{0j} + \sum_{k \in S_i} \phi_k \theta_{ik} + \varphi\theta_{ij} + \mu_m + \epsilon_{ijm}, \tag{7}$$

where S_i denotes patient i's choice set and μ_m are fixed effects for the primary practice HRR of the surgeon. Now, I only compare surgeons who practice within the same HRR, rendering the δ_{0j}'s interpretable.[16] Note also that the ϕ_k parameters are only estimated for the surgeons within a patient's HRR.

Directly estimating this regression is computationally burdensome, so I leverage the Frisch-Waugh-Lovell theorem to estimate the parameters and bootstrap to obtain standard errors. Because Equation 7 has a parameter for each surgeon in the dataset, of which there are more than 11,000, I cannot estimate it quickly via OLS. I therefore exploit the fact that the θ_{ik}'s are zero for surgeons outside a patient's HRR and use the Frisch-Waugh-Lovell theorem to obtain the estimates. I describe this estimation strategy in Appendix C.3. Finally, to correct the standard errors for the variation from the first-stage demand model, I modify a two-step score bootstrap approach to this setting in a procedure described in Appendix C.5 (Kline and Santos, 2012).

3.4 Identification

To aid the identification discussion, I first provide intuition for the identification of the three types of control function coefficients. Broadly, these coefficients in Equation 7 are identified by comparing patients who travel short versus far distances for a hip or knee replacement. Identification of κ is standard. If patients travel farther for j relative to other surgeons and j's patients have a lower

[15] For 8.1% of hip and knee replacements, surgeons do not perform a replacement in their assigned ZIP code. Thus, for patients who choose surgeons at these non-assigned ZIP codes, I assign the surgeon only to the ZIP code where the surgeon performs the replacement.

[16] Following the procedure described in Appendix C.1, I also use Empirical Bayes to shrink these parameters to the market-level mean to address measurement error.

probability of a complication, $\kappa < 0$. Identification of ϕ_k and φ is slightly less conventional. If, for instance, surgeon k treats unobservably less risky patients, then $\phi_k < 0$, since the unobserved reasons for choosing surgeon k (as captured in η_i and revealed through distance) are negatively correlated with the probability of a complication. For the same reason, if patients select their chosen surgeon j for a match-specific reason, as captured in η_{ij}, then $\varphi < 0$. Critically, if the relative distance patients travel to a surgeon is correlated with the choice determinants, such as surgeon quality or a patient's unobserved risk, then it carries information about these factors. In this case, controlling for δ_{0j} and the η_i's (implicitly through the θ_{ij}'s) would no longer fully capture these factors that matter for outcomes.

Causal identification of the returns to surgeon volume first requires a relevance assumption—distance must affect patient choice of surgeon. To assess this assumption, I show the predicted choice probability of a surgeon versus the distance to that surgeon for the Boston HRR in Appendix Figure C1. As evident in the figure, distance is a strong predictor of patient choice. The average choice probability falls by 71% when a patient is 0–5 miles away from a surgeon versus 5–10 miles away. This relationship generally holds across all markets as well, as the average coefficient on distance across markets implies that doubling the distance to a surgeon decreases the odds of choosing that surgeon by 52%, as shown in Appendix Figure C2. In part for this reason, distance is a commonly used instrument in the health care literature (Grabowski et al., 2013; Einav, Finkelstein, and Williams, 2016; Einav, Finkelstein, and Mahoney, 2022; Card, Fenizia, and Silver, 2023). This figure also supports the first-stage functional form assumption, as a logarithmic relationship approximates well the relationship between choice and distance.[17]

Causal identification also requires the exclusion restriction hold—distance can only affect health outcomes through the choice of surgeon. Formally, this assumption implies that:

$$\begin{aligned} &\mathbb{E}\left[\epsilon_{ijm} | d_i, X_i, \gamma_{t(i)}, \widehat{\delta}_{0j}, \mu_m, D_i = j\right] = 0 \\ \iff &\mathbb{E}\left[\xi_j + \psi_i + \lambda_{ij} | d_i, X_i, \gamma_{t(i)}, \widehat{\delta}_{0j}, \mu_m, D_i = j\right] = 0, \end{aligned} \tag{8}$$

where I have re-expressed ϵ_{ijm} using the error term decomposition from Equation 3, as well as the fact that $\mathbb{E}\left[\omega_{ijm} | d_i, X_i, \gamma_{t(i)}, \delta_{0j}, \mu_m, D_i = j\right] = 0$. This formulation formalizes the exclusion restriction—knowing the relative distance patients travel produces no new information about the average value of the unobserved factors affecting outcomes, once the value of the controls is known. In other words, conditional on the controls, unobserved surgeon quality, ξ_j; unobserved patient characteristics, ψ_i; and unobserved patient-surgeon match characteristics, λ_{ij}, cannot be correlated with the relative distance patients travel. Hence, the major threats to identification occur if surgeons practice closer to patients based on quality not captured in patient demand ξ_j, patients live closer to surgeons based on their unobservable risk characteristics ψ_i, or patients and surgeons co-locate based on their match, λ_{ij}.

[17] In Appendix Tables C1 and C2, I show more evidence that the estimates are robust to this parameterization. That is, for one representative HRR, I demonstrate that the estimates from the demand model used to construct the control function essentially do not change under alternative distance specifications.

Concretely, the primary threats to identification are the endogeneity of surgeon and patient location decisions.[18] First, similar to the main identification threat in Dingel et al. (2023), if higher exogenous quality surgeons practice relatively closer to patients for reasons other than patient demand, the returns to surgeon volume would be biased upward. While much work points to labor demand as a key driver of skill-based geographic sorting (Berry and Glaeser, 2005; Moretti, 2013; Diamond, 2016; Baum-Snow and Hartley, 2020), other work highlights different causal forces, such as amenities or the rising value of time (Couture and Handbury, 2020; Su, 2022). Recall, however, that only skill-based sorting within an HRR is a threat, as the surgeon HRR fixed effects eliminate any bias induced by skill-based sorting across HRRs. Second, patient locations may be related to their unobservable risk characteristics.[19] For example, less risky patients may live relatively closer to surgeons, which would bias upward the returns to surgeon volume.

To address the threat from endogenous surgeon location decisions, I conduct two primary robustness exercises, both of which yield results that are consistent with the exclusion restriction. In the first one, I introduce several controls to the main estimating equation and show that these controls do not affect the main estimate. Specifically, I control for the rank of a surgeon's medical school to address any selection on exogenous quality that is not captured with labor demand. I also add controls for HSA and ZIP code market size, such that I compare surgeons who choose to practice in the same-sized markets. I further add controls for the log distance between surgeons' medical school and their practice locations and an amenity index for the surgeon's practice location that I describe in Appendix C.6. Other work has demonstrated that both training locations and amenities are important drivers of spatial sorting for physicians and perhaps even differentially for higher exogenous quality physicians (Seifer, Vranizan, and Grumbach, 1995; Lee, 2010). In a second robustness exercise, I investigate whether high-quality surgeons move to larger markets within an HRR, in the spirit of Chandra et al. (2016). Among 521 surgeon movers, I find that higher quality surgeons do not move to larger markets, as shown in Appendix Figure C3.[20]

To address the threat from endogenous patient location decisions, I conduct a covariate balance test and introduce a specification with patient ZIP code fixed effects and find evidence consistent with the exclusion restriction. Conducting robustness exercises with patients is simpler than it is with surgeons, as I have significantly more covariates and many more patients, allowing for more granular levels of fixed effects. Therefore, I first conduct a balance test to examine how the first-stage distance coefficient changes with the addition of observable covariates (Altonji, Elder,

[18] Another threat is that distance has a mechanical effect on outcomes, as would be the case for an emergent procedure that requires immediate transport to the hospital. Nevertheless, this third threat is not quite as serious, since 94% of the hip and knee replacements in the sample are elective.

[19] Note that this threat encapsulates two concerns. First, conditional on undergoing a hip or knee replacement, patients who live differentially closer to surgeons may have different unobserved characteristics. Second, patients who are differentially close to surgeons may select into receiving a hip or knee replacement based on unobserved patient characteristics (Chandra and Staiger, 2007). According to Chandra and Staiger (2007) and consistent with the findings in Appendix Table B1, if anything, this selection would bias downward the returns to surgeon volume. The two robustness exercises I describe below address both concerns.

[20] According to the Current Population Survey (CPS) from 2006–2019, the top reasons physicians and surgeons move are for family or different housing.

and Taber, 2005; Oster, 2019; Einav, Finkelstein, and Mahoney, 2022; Mourot, 2024). In Appendix Figure C4, I show that estimated distance coefficients for the 306 HRRs are essentially identical with and without the patient covariates. This tight correlation indicates that the effect of distance on a patient's choice does not change with observable patient covariates, lending credence to the exclusion restriction. To further bolster this point, I introduce an alternative specification with fixed effects for a patient's ZIP code and show that the main estimate does not change.[21]

4 Results: The returns to surgeon volume and market size

Using differential distance as an instrument, I find that an increase in a surgeon's hip and knee volume causally affects patients' health outcomes. This result indicates an externality—a patient's choice of surgeon affects other patients' health outcomes through the surgeon's quality. Using this estimate, I attribute 28% of the benefits of market size to the returns to surgeon volume. I then provide suggestive evidence that learning-by-doing is the main driver of the returns to surgeon volume relationship and show robustness of the result.

4.1 Main results

Addressing selection using differential distance as an instrument, I show that doubling a surgeon's hip and knee volume reduces the probability of a complication by 11%. Focusing first on the OLS estimate in column (1) indicates that doubling a surgeon's hip and knee volume is associated with an 18% reduction in the probability of a complication. However, as discussed earlier, this coefficient may be biased upward or downward for several reasons. Thus, to obtain the main estimate, I introduce the differential distance instrument with the control function approach in column (2), as well as the measurement error correction procedure from Appendix C.4, and demonstrate that the effect size falls by 38%. In Appendix Table D3, I progressively add each of these terms and show that addressing selection on exogenous surgeon quality (controlling for δ_{0j}) and unobservable patient risk (controlling for the θ_{ik}'s) decreases the estimate, while the measurement error correction increases it. The magnitude is slightly smaller than those found in other papers using quasi-random variation, even those in similar orthopedic contexts (Hentschker and Mennicken, 2018; Fitzgerald, Gupta, and Schwab, 2026).

Exploring the types of selection that bias the OLS estimate, I find that selection on exogenous quality biases it upward. Note that in Table 2 column (2), the coefficient on the demand for exogenous surgeon quality, $\widehat{\delta}_{0j}$, is negative and statistically significant. This coefficient implies an increase in the demand for exogenous surgeon quality is associated with a decline in the probability of a complication, an observation that is also supported in Appendix Figure D1. Using the Boston HRR as an example, the magnitude implies that moving from the 25th percentile to the 75th percentile in the distribution of $\widehat{\delta}_{0j}$ corresponds to a 12% decline in the probability of a complication. Hence, the OLS coefficient is biased upward because patients are more likely to choose higher exogenous quality surgeons, higher exogenous quality surgeons are more likely

[21] Even though the finest geographic unit I observe is a ZIP code, I still have variation in differential distance because a patient's choice set also depends on when they undergo the replacement.

	(1) Complication	(2) Complication
Log volume $(\ln v_{ij})$	-0.0055	-0.0034
	(0.0003)	(0.0008)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$		-0.0046
		(0.0018)
Roy selection $(\theta_{ij}(j))$		-0.0002
		(0.0004)
N	696,620	696,620
Mean complication	0.031	0.031
Standard deviation surgeon selection term $(\widehat{\phi}_k)$		0.014
Risk covariates	✓	✓
Year fixed effects	✓	✓
Surgeon HRR fixed effects	X	✓
Measurement error correction	X	✓

Table 2: Estimates of the returns to surgeon volume

Notes: This table shows the estimates from estimating Equation 7. The outcome is a binary indicator equal to one if a patient has a complication. All specifications are estimated using a linear probability model. The specification in column (1) provides the OLS estimate. The specification in column (2) instruments for surgeon volume using differential distance and adjusts the coefficient on log volume for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5 for column (2).

to practice in larger markets, or both. This result echoes other work that indicates that patient demand plays an important role in allocating patients to health care providers (Chandra et al., 2016). From a policy perspective, it suggests that, even without a causal effect of surgeon volume on outcomes, policies that re-allocate patients to surgeons with higher hip and knee volume may generate welfare improvements through the surgeon's exogenous quality.

I also find suggestive evidence that unobservably healthier patients choose higher hip and knee volume surgeons. Returning to column (2) in Table 2, the standard deviation of the surgeon-specific selection term coefficients, $\widehat{\phi}_k$, as well as the distribution shown in Appendix Figure D2, suggests sizable dispersion across surgeons in these selection parameters. That is, some surgeons treat unobservably riskier patients while others treat unobservably healthier patients. To better demonstrate how this selection affects the OLS estimate, I show in Appendix Figure D3 that surgeons with higher average annual hip and knee volume have lower values of $\widehat{\phi}_k$, indicating that unobservably less risky patients choose higher hip and knee volume surgeons. This selection thus biases the OLS estimate upward, a result that is consistent with the decline in the coefficient on log volume from column (1) to column (2) in Appendix Table B2 when adding observable patient covariates. One potential explanation for this selection mechanism is that healthier individuals have higher income and thus are less sensitive to traveling far or have better information (Dingel et al., 2023; Mourot, 2024).

Finally, I find that patients do not appear to select on an idiosyncratic match with a surgeon. Once again returning to column (2) in Table 2, note that the coefficient on the Roy selection con-

trol, $\theta_{ij}(j)$, is negative but quite small and statistically insignificant. While the negative coefficient would be suggestive of Roy selection, the small magnitude and lack of statistical significance indicate that such match-specific selection is likely unimportant in this setting. It therefore does not meaningfully bias the OLS estimate. Roy selection is also negligible in other health care settings (Einav, Finkelstein, and Mahoney, 2022; Mourot, 2024).

Using the causal returns to surgeon volume, I can then attribute 28% of the benefits of market size to the returns to surgeon volume. Because the returns to surgeon volume relationship also underscores an externality, this result may be especially relevant for designing place-based policy in health care. It also provides an empirical magnitude to previous theoretical contributions that have demonstrated that learning and specialization are microfoundations underlying the benefits of agglomeration (Duranton and Puga, 2004).

4.2 Mechanisms

I first show that dynamic economies of scale, like learning-by-doing, contribute meaningfully to the returns to surgeon volume relationship. Following Gaynor, Seider, and Vogt (2005), I introduce lagged surgeon hip and knee volume into the estimating equation to help disentangle static and dynamic economies of scale.[22] The coefficient on contemporaneous volume captures both static scale economies, such as higher quality team members, and dynamic learning-by-doing, if skills depreciate (Benkard, 2000; Fitzgerald, Gupta, and Schwab, 2026). Meanwhile, the coefficient on lagged volume captures dynamic differences, such as learning-by-doing, on-the-job experience or changes in team-specific human capital, which others show to be important in various settings (Irwin and Klenow, 1994; Levitt, List, and Syverson, 2013; Chen, 2021; Dinerstein, Megalokonomou, and Yannelis, 2022). Table 3 shows that the coefficients on both contemporaneous and lagged volume are economically meaningful and statistically significant at conventional levels. Doubling contemporaneous hip and knee volume corresponds to a 9% decline in the probability of a complication, whereas the effect is about 4% for lagged volume. While the coefficient on contemporaneous volume is larger, I cannot statistically reject that these estimates are equal. Hence, dynamic economies of scale at least partially drive the returns to surgeon volume relationship, and either static scale economies are also important or skills depreciate over time.

Exploring specific mechanisms, I find that learning-by-doing, as opposed to static or other dynamic scale economies, appears to be the primary driver of the returns to surgeon volume relationship. I first examine two specific static scale economies—hospital scale and team quality, as proxied by the anesthesiologist's hip and knee volume—that others have shown to be important for outcomes in similar settings (Birkmeyer et al., 2003; Hallet et al., 2021). Table 4 demonstrates that adding hospital hip and knee volume and the anesthesiologist's hip and knee volume to the

[22]Motivated by the fact that the descriptive relationship between the probability of a complication and lagged measures of volume dies out after just one lag, as shown in Appendix Figure A1, I just include one lag. To avoid the computational burden of re-estimating the demand model, I omit lagged volume from the demand model. However, for one representative HRR, I show that the estimates from the demand model used in the main estimating equation essentially do not change when introducing lagged surgeon hip and knee volume into the demand model in Appendix Tables D1 and D2.

	(1)	(2)
Model	Control function	Control function
Outcome	Complication	Complication
Log volume $(\ln v_{ij})$	-0.0036	-0.0028
	(0.0009)	(0.0008)
Log lagged volume $\left(\ln v_{ij}^{lag}\right)$		-0.0014
		(0.0006)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$	-0.0046	-0.0042
	(0.0020)	(0.0021)
Roy selection $(\theta_{ij}(j))$	-0.0002	-0.0003
	(0.0005)	(0.0005)
N	639,979	639,979
Mean complication	0.032	0.032
Standard deviation surgeon selection term $(\widehat{\phi}_k)$	0.017	0.017
Risk covariates	✓	✓
Year fixed effects	✓	✓
Surgeon HRR fixed effects	✓	✓
Measurement error correction	✓	✓

Table 3: Decomposing the returns to surgeon volume into static and dynamic economies of scale

Notes: This table shows the estimates from estimating Equation 7. The outcome is a binary indicator equal to one if a patient has a complication. Both specifications are estimated using a linear probability model. The specification in column (1) omits log lagged hip and knee volume, whereas the specification in column (2) adds it. The coefficients on log volume are adjusted for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5.

main specification in Equation 7 only reduces the returns to surgeon volume by a statistically insignificant 12%. Thus, static economies of scale do not appear to meaningfully mediate this relationship, a finding that is consistent with Birkmeyer et al. (2003). Focusing on a specific dynamic economy of scale, I also show in the same table that changes in team-specific human capital, as proxied by whether the surgeon and anesthesiologist have worked together in the previous 365 days, do not drive this relationship, despite the importance of this coordination for patient outcomes (Chen, 2021). Finally, in Appendix Table D4, I show in a much smaller sample that choice of procedural technique does not mediate the relationship, even though it has a statistically significant association with better outcomes.[23] While these mechanisms are not necessarily exhaustive or causal, the fact that the effect loads on surgeon volume in these specifications suggests that learning-by-doing drives the returns to surgeon volume relationship. Coupled with the lagged specification, these results together then indicate that skills depreciate over time, which is consistent with other work on surgeon learning and forgetting (Lundborg et al., 2021; Fitzgerald, Gupta,

[23]The specific procedural technique I use is if the replacement is cemented. Cemented fixation is generally preferred for older patients, as it typically offers a lower risk of revision, while cementless fixation is more common in younger patients. Because I can only observe this technique in the last four years of my dataset, the sample size falls substantially and the main effect becomes statistically insignificant, although the same qualitative patterns emerge.

and Schwab, 2026).

	(1) Complication	(2) Complication	(3) Complication	(4) Complication
Log surgeon volume $\left(\ln v_{ij}\right)$	-0.0033	-0.0032	-0.0029	-0.0028
	(0.0008)	(0.0008)	(0.0009)	(0.0009)
Log hospital volume (excl. focal surgeon)		-0.0005	-0.0002	-0.0002
		(0.0003)	(0.0004)	(0.0004)
Log anesthesiologist volume			-0.0012	-0.0012
			(0.0003)	(0.0004)
Surgeon-anesthesiologist prior experience				-0.0002
				(0.0005)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$	-0.0047	-0.0045	-0.0045	-0.0045
	(0.0020)	(0.0020)	(0.0020)	(0.0020)
Roy selection $\left(\theta_{ij}(j)\right)$	0.0003	0.0005	0.0004	0.0004
	(0.0006)	(0.0007)	(0.0007)	(0.0007)
N	519,964	519,964	519,964	519,964
Mean complication	0.030	0.030	0.030	0.030
Standard deviation surgeon selection term $\left(\widehat{\phi}_k\right)$	0.032	0.032	0.032	0.032
Risk covariates	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓
Surgeon HRR fixed effects	✓	✓	✓	✓
Measurement error correction	✓	✓	✓	✓

Table 4: Examining specific mechanisms underlying the returns to surgeon volume

Notes: This table shows the estimates from estimating Equation 7. The outcome is a binary indicator equal to one if a patient has a complication. All specifications are estimated using a linear probability model. Column (2) adds a control for the hospital's log volume (less the surgeon's volume). Column (3) adds a control for the anesthesiologist's log volume. Column (4) adds a binary control equal to one if the surgeon and anesthesiologist have worked together in the previous 365 days. The coefficients on log volume are adjusted for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5.

4.3 Robustness and alternative specifications

To address the potential bias from endogenous surgeon and patient locations, I show that these results do not change when I add various surgeon and market-level controls or add fixed effects for the ZIP codes where patients live. Recall that if higher exogenous quality surgeons sort into larger markets based on characteristics other than patient demand, the returns to surgeon volume estimate would be biased upward. Thus, comparing column (1) to column (2) in Appendix Table D5, I show that the estimate of the returns to surgeon volume does not change when I control for the rank of a surgeon's medical school, HSA market size, ZIP code market size, the distance between the surgeon's practice location and medical school, and the amenity index of the ZIP code where the surgeon operates. These results therefore lend support to the assumption that surgeon locations are conditionally exogenous. In Appendix Table D6, I also show that adding fixed effects for a patient's ZIP code does not noticeably change the results, suggesting that patient locations are conditionally exogenous.

In additional specifications, I introduce hospital and surgeon fixed effects into the main estimating equation both to alleviate remaining identification concerns and further explore mech-

anisms. Appendix Table D7 demonstrates that hospital and surgeon fixed effects decrease the returns to surgeon volume relationship by 22% and 28%, respectively, but the estimates fall within the 95% confidence interval of the estimate without the fixed effects and are still statistically significant at conventional levels.[24] Interpreting these estimates is somewhat challenging. On one hand, the fixed effects address any remaining endogeneity concerns due to surgeons' endogenous location decisions, as they rely on within-hospital or within-surgeon comparisons. On the other hand, they also provide further insight on the mechanisms underlying the relationship. For instance, the fixed effects eliminate cross-hospital or cross-surgeon differences in nurse quality. Additionally, they also exacerbate the attenuation bias due to measurement error in volume (Griliches and Hausman, 1986; Lundborg et al., 2021). Finally, a conceptual issue arises in interpreting the specification with surgeon fixed effects if annual hip and knee volume is considered a proxy for cumulative hip and knee volume.

To probe the robustness of my results, I show that the returns to surgeon volume relationship is consistent across alternative specifications.[25] In Appendix Table D8, I show that the results are robust to eliminating surgeons with zero hip and knee volume and not adding one to volume. In a similar manner, I show the estimates of the returns to surgeon volume using volume quintiles in Appendix Table D9, which yields similar results. Additionally, I separately identify the effects for hip replacements and knee replacements in Appendix Table D10 and find a slightly larger effect for knee replacements than for hip replacements, although the difference is not statistically significant. Finally, in Appendix Table D11, I examine how spillovers across other procedures affect the estimate by controlling for the volume of other common orthopedic surgical procedures but do not find that these controls meaningfully affect the estimate.

5 Policy implications

To understand the policy implications of the returns to surgeon volume externality, I use the demand model and the causal estimate to evaluate the welfare consequences of a first-best policy and three prominent feasible policies. Failing to incorporate the returns to surgeon volume relationship substantially understates the welfare effects of the feasible policies. A minimum volume standard increases welfare by more than subsidizing transportation or moving surgeons to shortage areas, yet it falls far short of a first-best policy.

5.1 Defining and interpreting welfare

I define welfare as consumer surplus less fiscal costs and incorporate the returns to surgeon volume externality into both consumer surplus and fiscal costs. More precisely, using m to denote

[24] In practice, for ease of computation, I treat these fixed effects as local parameters, implying that I actually add hospital- and surgeon-by-patient market fixed effects.

[25] As with the specification that includes lagged volume, I do not change the demand model under these alternative functional forms to avoid the computational burden of re-estimating the demand model. However, for one representative HRR, I show that the estimates from the demand model used in the main estimating equation essentially do not change under these alternative demand model functional forms in Appendix Tables D1 and D2.

HRRs, I define welfare, $\mathcal{W}$, as:

$$\begin{aligned}\mathcal{W} &= \sum_m s_m \mathcal{W}_m \\ &= \sum_m s_m \left(CS_m - c \times N_{comp,m} - G_m \right), \end{aligned} \tag{9}$$

where s_m is the share of total patients in market m, CS is consumer surplus as defined in Appendix E.1, c is the dollar value of a complication,[26] N_{comp} is the number of complications,[27] and G is the cost of implementing the policy. While this definition captures the key elements of patient and government welfare, it does not consider producer surplus, nor any general equilibrium effects, as these components are beyond the scope of this paper.

This welfare function embeds two externalities. The first is that a patient's choice of surgeon affects other patients' outcomes. I incorporate this externality both in consumer surplus, as patient utility depends on a surgeon's hip and knee volume through the demand model, and in the fiscal cost of complications, as the number of complications depends on the quality of surgeons and thus on surgeon volume. In practice, therefore, I compute equilibrium outcomes allowing for feedback loops between a surgeon's volume and patients' probability of choosing that surgeon.[28] Second, I incorporate the fiscal cost of complications because patients do not face full financial responsibility for their care with insurance, and the fiscal cost of a complication to the Medicare program is high (Yi et al., 2015).[29] With these externalities, policies change welfare if they change patient choice. The new allocation of patients to surgeons not only affects those who switch surgeons but also the patients who do not switch, as the quality of their surgeon changes. Hence, both the direction and magnitude of the welfare effects of the policies I evaluate are theoretically ambiguous.

Finally, since latent choice constraints, such as available capacity or imperfect information, may affect the demand estimates and hence welfare, I implement a capacity constraint and hold fixed all other constraints under alternative policy regimes. Recall from Section 3.1 that δ_{0j} and δ_1 may be correlated with other surgeon characteristics that affect patient demand, such as available capacity. To address this issue, I first impose a volume constraint at the 99th percentile of observed hip and knee volume. If a surgeon's volume in the 365 days prior to the patient's surgery date exceeds the constraint, I omit the surgeon from the patient's choice set. Second, I assume all other choice constraints, such as imperfect information, are fixed across alternative policy regimes.[30]

[26] To calibrate the fiscal cost c, I rely on Yi et al. (2015) who estimate the additional Medicare payments in the four years following a hip or knee replacement to be \$53,470 higher for patients with an infection than without one (\$65,442 after converting to 2012 dollars). While my complication measure includes adverse consequences other than infections, infections are the most common. If anything, I likely understate the true cost, as death occurs in 7% of complications.

[27] I estimate N_{comp} using surgeons' volume and their average risk-adjusted complication rate that I shrink to the mean using the Empirical Bayes procedure described in Appendix C.1.

[28] The parameters δ_1 from Equation 2 govern the magnitude of the change from the feedback loops, so I show the distribution of δ_1 in Appendix Figure E3. To limit the influence of noisier estimates of δ_1 in smaller markets, I shrink them to the national mean using the Empirical Bayes procedure described in Appendix C.1.

[29] Other work also suggests that higher volume surgeons have lower costs of care (Sauder et al., 2026). I do not incorporate this additional fiscal externality as I do not observe costs.

[30] This assumption implies that the total time surgeons spend working is constant across policy regimes, such that

5.2 Calculating a first-best policy

To benchmark the three feasible policies and inform the targeting of policy, I compute the welfare gain from a first-best policy in which a social planner optimally assigns patients to surgeons. I first consider how the social planner would assign patients in two simplified cases to provide intuition for this exercise. Suppose for now that patients face no travel costs and surgeons cannot be capacity constrained. In this case, the social planner would assign all patients to the highest exogenous quality surgeon, which would result in a further welfare increase through the returns to surgeon volume externality. However, with capacity constraints, assigning more patients to one surgeon helps patients who choose the now higher volume surgeon but worsens outcomes for those who remain with the now lower volume surgeons. Meanwhile, with non-zero travel costs, the solution depends on the relative importance of the externality versus the travel costs, as I show in a simplified setting in Appendix E.2.

I generalize this intuition by introducing surgeon-specific Pigouvian taxes or subsidies into the demand model that the social planner manipulates to maximize welfare and achieve the first-best allocation. Formally, I modify the demand model from Equation 1 as follows:

$$u_{ij} = \delta_j(v_{ij}) - \tau \ln d_{ij} + \pi_j + \eta_{ij}, \tag{10}$$

where all else is defined as before and π_j denotes the tax (or subsidy) for surgeon j. The social planner manipulates these parameters to steer patients to certain surgeons based on the welfare function. Intuitively, these terms can be thought of as surgeon-specific co-payments that a savvy private insurer might set. Because they function as pure transfers and exactly cancel out with fiscal spending or revenue, they are not used when calculating welfare. With this formulation, solving for the vector of tax parameters in the demand model allows me to recover welfare under the first-best policy. To solve for them, I use bound optimization by quadratic approximation, a non-linear optimization algorithm that finds the local maximum of reasonably well-behaved functions without derivatives.[31] Because, however, this optimization is still quite computationally burdensome, I restrict to HRRs with fewer than fifty-six surgeons, which is 75% of HRRs.[32]

I show first that the social planner consolidates care among high-quality surgeons by introducing large subsidies for these surgeons but that the subsidies are much lower with the returns to surgeon volume externality. Figure 2 demonstrates that the planner's optimal taxes can be quite large, as they are a sizable share of the average $1,500 that Medicare FFS pays a surgeon for a hip or knee replacement. As patients respond to these large taxes, care consolidates among a

they substitute to and from other procedures or work activities when their hip and knee volume changes. Appendix Figure E2 presents evidence consistent with this assumption, as surgeons who perform more hip and knee replacements allocate a larger share of their work, as measured in relative value units (RVUs), to hip and knee replacements.

[31] While I cannot prove that the optimal tax assignment I solve for is the unique solution, I numerically show that randomly drawing different starting tax vectors for one HRR generates qualitatively similar welfare conclusions both without and with the returns to surgeon volume externality in Appendix Figures E4 and E5.

[32] Plotting the magnitude of the welfare effect versus the number of surgeons in the market exhibits no significant relationship, as shown in Appendix Figure E6.

much smaller set of surgeons, as shown in Appendix Figure E7. Hence, averaging over the results with and without the externality, the Herfindahl Hirschman Index (HHI) increases from 0.08 to 0.62. Nevertheless, with the externality, the tax parameters are substantially smaller in magnitude because the market amplifies the changes in surgeon volume that the taxes induce, as patient demand responds to volume.

Figure 2 also shows that the tax parameters exhibit a strong negative correlation with a surgeon's complication rate but that this relationship is weaker with the externality. That is, the social planner subsidizes surgeons with low complication rates and taxes those with high ones. The slope of the relationship between the tax parameters and surgeon quality also falls with the externality, implying that the increase in the HHI is actually 41% smaller with the externality than without it. This result reflects a fundamental tradeoff with the externality—funneling patients to high-quality surgeons is welfare-increasing for patients who choose high-quality surgeons but welfare-decreasing for those who choose other surgeons.

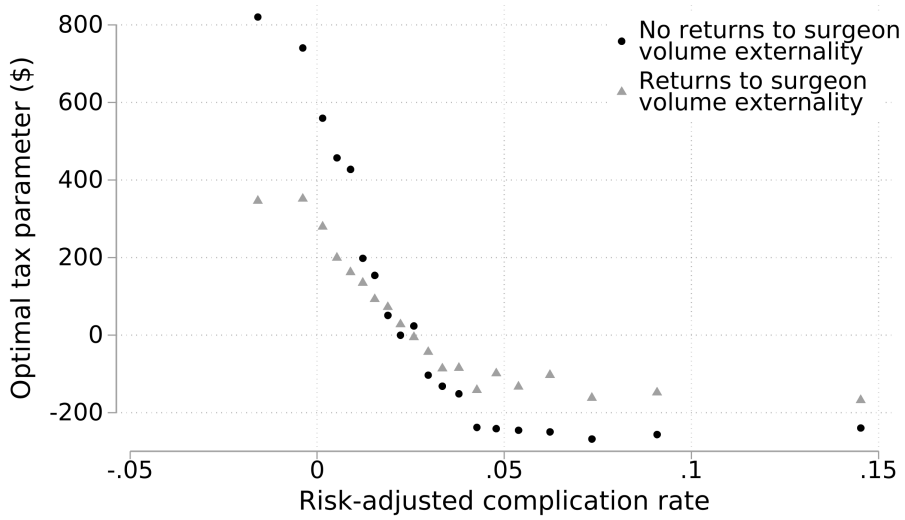


Figure 2: Relationship between optimal tax parameter and complication rate

Notes: This figure shows the relationship between the optimal tax parameter for a surgeon, π_j, and the surgeon's risk-adjusted complication rate. The optimal tax parameter has been converted into dollars following the method in Appendix E.1. The risk-adjusted complication rate is the surgeon fixed effect after partialling out the risk covariates listed in Appendix Table A2 and has been shrunk to the mean using Empirical Bayes, as described in Appendix C.1. The relationship with the black dots shows this relationship without incorporating the returns to surgeon volume externality (i.e., setting $\delta_1 = 0$ from Equation 1 and $\beta_1 = 0$ from Equation 7). The relationship with gray triangles shows the relationship when incorporating the returns to surgeon volume externality. The underlying unit of observation is a surgeon-HRR that has been grouped into twenty equal-sized bins based on a surgeon's risk-adjusted complication rate after residualizing out HRR fixed effects.

Finally, I show that the welfare effects are large under this first-best policy both with and without the externality. Appendix Table E1 demonstrates that the change in welfare per person under this policy hovers around $2,000, which actually exceeds the average amount Medicare FFS pays to surgeons for a replacement. Notably, the externality slightly decreases the welfare effect, which

reflects the cost of consolidating care on those who remain with the lower volume surgeons. Increases in fiscal savings drive these large welfare effects, as the social planner re-allocates patients to higher exogenous quality surgeons. Consumer surplus actually declines, although the magnitude is trivial compared to the change in fiscal savings. In Appendix Figure E8, I also show that the welfare effects for rural and urban patients are quite similar under this optimal policy both with and without the externality.

5.3 Introducing the feasible counterfactual policies

For the feasible policies, I first consider a minimum volume standard in which surgeons who fall below a volume threshold are not permitted to perform hip and knee replacements. Minimum volume standards are especially common at the state level for stroke and cardiac care, and some prominent interest groups, such as the Leapfrog Group, advocate for them more broadly.[33] Theoretically, the welfare effect of this policy is unclear. On one hand, patients may receive higher quality care both if they now choose higher exogenous quality surgeons and through the returns to surgeon volume externality. On the other hand, this policy will reduce consumer surplus through a reduction in the number of choices and an increase in travel distance. To implement this policy, I remove any surgeon from a patient's choice set whose hip and knee volume does not exceed twenty. While the number of choices in the average patient's choice set falls by 48%, these low hip and knee volume surgeons only perform 12% of hip and knee replacements. I set G_m, the cost of implementing the policy, to be zero, as there is no direct cost of implementation.

I also consider a policy that subsidizes transportation for patients. While these subsidies do not currently exist within the Medicare FFS program, 44% of Medicare Advantage plans offer some type of coverage for non-emergency medical transportation as of 2024 (Shen et al., 2024).[34] On one hand, the subsidy may incentivize patients to travel to further centralize care. On the other hand, without the returns to surgeon volume externality and the fiscal externality, this subsidy is purely distortionary, since the patients it induces to travel farther value the farther surgeon by less than the full cost of the trip. Hence, the fiscal cost from the subsidy may exceed the benefit of the policy if the returns to surgeon volume externality is relatively small. In practice, I subsidize 30% of patients' transportation costs, so I cut τ from the demand model in Equation 1 by 30%. Meanwhile, I calibrate G_m, the fiscal cost of implementing the policy, using the wage in the patient's county plus the dollar cost of driving, as described in Appendix E.1.[35] Finally, to isolate welfare changes resulting from the reallocation of patients to surgeons and not from the mechani-

[33] Minimum volume standards are also quite similar to network restrictions that introduce strong financial incentives for patients to choose high volume surgeons. While Medicare FFS has no such restrictions, they are quite common in Medicare Advantage. Some Medicare Advantage insurers even require networks to contain high volume specialists.

[34] This coverage for transportation often reimburses patients for transportation within certain geographic bounds, and it may restrict the number of trips in a policy year or require some cost-sharing. While I do not consider these more complicated policy designs, the welfare analysis still provides insight into the effects of these supplemental insurance benefits.

[35] Intuitively, this value might reflect the distance and time cost to patients if family members or friends drive them. Note I also assume that the cost of raising taxes to finance this policy is zero, as the government can raise revenues with a non-distortionary lump sum tax.

cal increase in welfare from surgeons being effectively closer, I compute the welfare change using a compensated change in consumer surplus, as described in Appendix E.3.

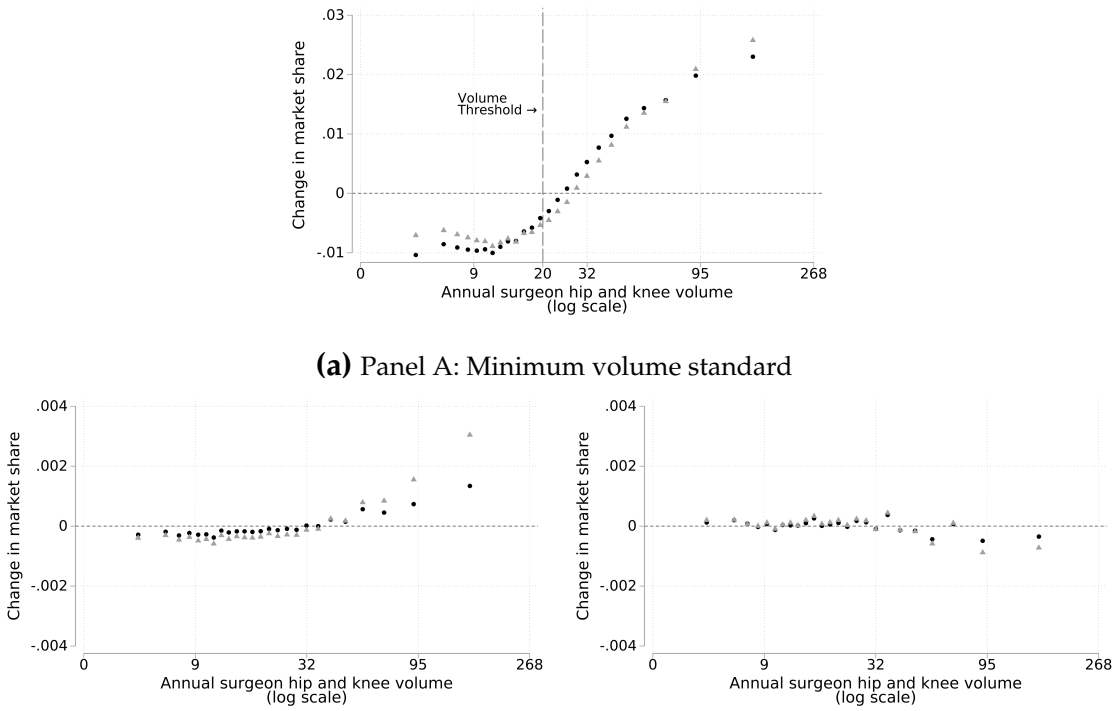
Finally, I consider a policy in which I move surgeons to government-designated shortage areas. Congress has enacted various similar policies to incentivize surgeons to move to shortage areas, such as educational loan repayments or increased Medicare payments. Despite their prevalence, the welfare effects of such initiatives are ambiguous even for the patients within shortage areas. While the presence of more surgeons decreases travel distance, it also spreads volume over a wider number of surgeons, potentially resulting in lower volume and lower quality surgeons. Complicating this analysis is the fact that the ultimate welfare results depend on what happens to patients in areas where the surgeons move from, which surgeons move, and the fiscal cost of subsidizing surgeons to move. I implement this policy by randomly moving surgeons into hospitals located in shortage areas and describe the implementation in more detail in Appendix E.4. I set the cost of implementing the policy, G_m, equal to zero, so this welfare analysis can be considered an upper bound on the true effect. In total, I move 1,350 surgeons, which is 12% of all the surgeons in my sample, to the same number of shortage areas within 68% of the HRRs.

5.4 The effect of the feasible policies on concentration

To preview the welfare results, I show that the minimum volume standard changes concentration among high-volume surgeons more than the other two feasible policies and that the returns to surgeon volume externality amplifies the changes in concentration for all the policies. Figure 3 first demonstrates that the minimum volume standard and the transportation subsidy increase concentration among high-volume surgeons, whereas moving surgeons to shortage areas very slightly decreases it. Notably, however, the magnitude of this change is almost an order of magnitude larger for the minimum volume standard than it is for the other two policies, as the minimum volume standard mechanically induces dramatic changes in patient choice. Additionally, the figure shows that the externality amplifies these changes in concentration. For example, for the minimum volume standard and transportation subsidy, the increase in the market share of surgeons above the 95th percentile of volume is 10% and 124% greater with the externality, respectively. Meanwhile, when moving surgeons to shortage areas, the decrease in market share among these same surgeons is 94% larger with the externality.

5.5 Welfare effects of feasible policies

Examining the minimum volume standard first, I show it increases welfare both as patients choose higher volume surgeons and through the returns to surgeon volume externality. Table 5a demonstrates that the policy increases welfare by $125 per patient without the externality. As the table shows, fiscal savings due to a lower complication rate drive this result. Echoing the results from the control function estimation shown earlier in Table 2, when surgeons with low hip and knee volume are no longer permitted to operate, patients re-sort to higher hip and knee volume surgeons, who also have higher exogenous quality. These fiscal savings vastly outweigh the (mechanical) decline in consumer surplus resulting from less choice and longer travel distances. Once



(a) Panel A: Minimum volume standard

(b) Panel B: Subsidizing transportation

(c) Panel C: Moving surgeons

Figure 3: Relationship between change in market share and surgeon hip and knee volume

Notes: Each of these figures shows the relationship between the change in a surgeon's market share and the surgeon's average log volume after implementing a minimum volume standard (Figure 3a), a 30% subsidy for patient transportation (Figure 3b) and moving surgeons to government-designated shortage areas (Figure 3c). The relationship with the black dots shows this relationship without incorporating the returns to surgeon volume externality (i.e., setting $\delta_1 = 0$ from Equation 1). The relationship with gray triangles shows the relationship when incorporating the returns to surgeon volume externality. The underlying unit of observation is a surgeon-HRR that has been binned into twenty-five equal-sized bins based on a surgeon's average log hip and knee volume after residualizing out HRR fixed effects. For better visualization, the top figure is on a different scale than the bottom two figures.

I incorporate the returns to surgeon volume externality, the welfare effect of the minimum volume standard increases by 31% up to $164 per patient. Increases in both consumer surplus and fiscal savings drive this result because higher surgeon volume now increases patient utility and improves outcomes. Nevertheless, even with the externality, this effect is only 9% of the first-best benchmark, which is perhaps unsurprising given that the minimum volume standard imposes a coarsely targeted infinite tax on low-volume surgeons and no tax on all others.

	No returns to surgeon volume externality	Returns to surgeon volume externality
Panel A: Minimum volume standard		
$\Delta\mathcal{W}$ / person ($)	125.38	163.72
Δ (CS - G) / person ($)	-20.80	-12.94
Δ fiscal savings / person ($)	146.18	176.66
Panel B: Subsidizing transportation		
$\Delta\mathcal{W}$ / person ($)	-6.80	4.33
Δ (CS - G) / person ($)	-20.63	-20.63
Δ fiscal savings / person ($)	13.83	24.96
Panel C: Moving surgeons		
$\Delta\mathcal{W}$ / person ($)	2.27	7.09
Δ (CS - G) / person ($)	1.47	1.58
Δ fiscal savings / person ($)	0.79	5.51

Table 5: Welfare effects of feasible policies

Notes: This table shows the welfare effects of implementing a minimum volume standard (Table 5a), a 30% transportation subsidy (Table 5b), and moving surgeons to government-designated shortage areas (Table 5c). The first row of each panel shows the average change in per-person welfare in dollars, as calculated using Equation 9. The second and third rows show the average change in the (per-person) sub-components of welfare: consumer surplus less fiscal spending on policy implementation and fiscal savings from reduced complications, respectively. The first column shows the effect without the returns to surgeon volume externality (i.e., setting $\delta_1 = 0$ from Equation 1 and $\beta_1 = 0$ from Equation 7), while the second column incorporates the returns to surgeon volume externality.

Next, I show that a 30% transportation subsidy decreases welfare without the returns to surgeon volume externality but increases it with it. The first column of Table 5b indicates that the subsidy decreases welfare by approximately $7 per patient without the externality. A decrease in consumer surplus less the fiscal costs of policy implementation is responsible for this decline. The subsidy does slightly increase fiscal savings as patients choose higher exogenous quality surgeons, but these savings are insufficient to generate a positive welfare effect. Then, in the second

column of Table 5b, I show that incorporating the externality actually flips the sign of the welfare effect. The primary driver of this sign reversal is an almost two times increase in fiscal savings as care becomes more consolidated and surgeon quality improves. Despite this increase in welfare, the effect is still quite small. It is less than 3% of the gain from the minimum volume standard and less than 1% of the first-best benchmark.

I now show that moving surgeons to shortage areas generates a small increase in welfare without the returns to surgeon volume externality and, surprisingly, an even larger increase with it. The first column of Table 5c demonstrates that moving surgeons to shortage areas increases per-patient welfare by about $2 without the externality, as reduced travel increases consumer surplus. With the externality, the welfare effect increases to almost $7 per patient, since fiscal savings increase. While this somewhat surprising result masks substantial heterogeneity, the intuition is that surgeons in movers' origin locations experience a 10% increase in volume that increases welfare and more than offsets welfare declines from surgeons losing volume elsewhere, including the movers. Similar to the transportation subsidy, the effect of this policy is quite small, as it is only 4% of the minimum volume standard and less than 1% of the first-best policy. Even this effect, though, is an upper bound, as it does not incorporate the fiscal cost of moving surgeons. In fact, the cost of moving an individual surgeon must not exceed $1,300 annually per surgeon for this effect to remain positive, which is a trivial amount compared to orthopedic surgeons' average annual income of almost $800,000 (Gottlieb et al., 2023).

5.6 Welfare effects of feasible policies for rural versus urban patients

Examining heterogeneity in the welfare effects of the minimum volume standard, I show that it generates similar welfare effects for rural and urban patients but that rural patients bear more of the costs of the policy. As shown in Figure 4, the welfare effects of the minimum volume standard are roughly the same for rural and urban patients both with and without the externality. Intuitively, rural patients are now more likely to choose surgeons in urban markets, increasing welfare for both rural and urban patients. In fact, the share of hip and knee replacements performed in urban markets increases by 4%. Nevertheless, rural patients bear more of the cost of the policy. As shown in Appendix Figure E9, rural patients experience declines in consumer surplus that are approximately double what urban patients experience because they travel 7% farther but urban patients travel less. The greater decline in consumer surplus for rural patients is offset by a larger increase in fiscal savings, as these patients now choose higher exogenous quality surgeons.

I also show in Figure 4 that while the returns to surgeon volume externality increases the welfare effect of the transportation subsidy for both rural and urban patients, the welfare effect is positive for rural patients but negative for urban patients. The externality increases welfare for both rural and urban patients because rural patients use the subsidy to travel to larger markets, which increases the share of hip and knee replacements in urban markets by 5% and improves outcomes for both rural and urban patients. Meanwhile, the effect is positive for rural patients but negative for urban patients because rural patients use the subsidy to travel to surgeons with higher hip and knee volume and thus higher exogenous quality, which generates larger fiscal savings, as

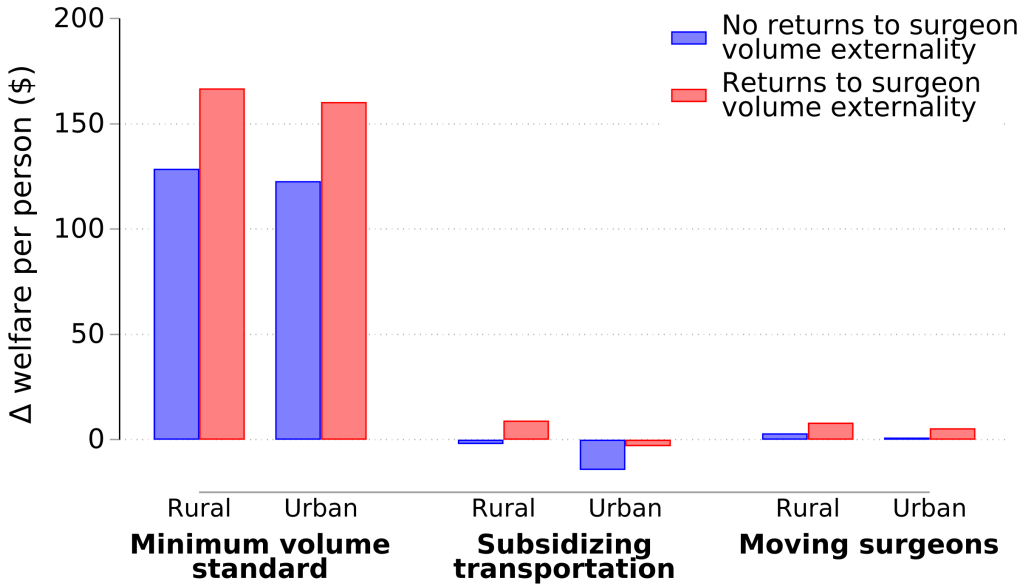


Figure 4: Welfare effects of feasible policies for rural versus urban patients

Notes: This figure shows the welfare effect of implementing a minimum volume standard, a 30% transportation subsidy, and moving surgeons to government-designated shortage areas for both rural and urban patients. Welfare is calculated in dollars using Equation 9. Rural patients are defined as those having ZIP codes of residence outside a "city center," where a "city center" is the HSA with the largest market size within an HRR. Urban patients are defined as those having ZIP codes of residence within a "city center." In 3 of the 306 HRRs, patients only live in one of the HSAs within the HRR, so these HRRs are omitted from this analysis. The blue bars show the effect without incorporating the returns to surgeon volume externality (i.e., setting $\delta_1 = 0$ from Equation 1 and $\beta_1 = 0$ from Equation 7), while the red bars incorporate this externality.

shown in Appendix Figure E10. In fact, the average hip and knee volume of the surgeons that operate on rural patients increases by 8% with the subsidy, as opposed to an increase of just 4% for urban patients. Nevertheless, even the welfare effect for rural patients is more than an order of magnitude smaller than that of the minimum volume standard or the first-best policy.

Finally, Figure 4 demonstrates that the returns to surgeon volume increase the welfare effect for both rural and urban patients but the effect is larger for rural patients. For rural patients, moving surgeons away from rural areas to urban areas actually drives the gains with the externality. In fact, in HRRs with only rural shortage areas, the externality decreases welfare for rural patients because moving surgeons experience an 11% decline in volume. For urban patients, though, the externality increases welfare regardless of whether surgeons enter or leave urban areas. Entering movers gain enough volume to outweigh the small, dispersed losses among the many other urban surgeons, while departing movers concentrate care among the remaining surgeons. Meanwhile, as shown in Appendix Figure E11, differences in fiscal savings primarily generate larger welfare effects for rural patients than for urban patients. Since moving surgeons are randomly chosen and rural surgeons on average have worse exogenous quality, fiscal savings increase more for rural patients. Still, though, as with subsidizing transportation, even the welfare effect for rural patients is only 4% of that for the minimum volume standard and less than 1% of that for the first-best policy.

6 Conclusion

While people in larger markets are more productive, relatively little is known about what drives these patterns, despite the importance of understanding these mechanisms for designing policy. In this paper, I first show that market size matters in health care. I then investigate the role of one potential mechanism–the returns to surgeon volume–in generating these benefits of market size. Using the differential distance between patients and surgeons as an instrument, I causally identify the returns to surgeon volume and attribute 28% of the benefits of market size to this mechanism. This returns to surgeon volume relationship highlights an externality–patients' choice of surgeon affects other patients' outcomes through the quality of the surgeon. To understand the policy implications of these findings, I incorporate this returns to surgeon volume externality into a demand model and evaluate the welfare consequences of a first-best policy and three feasible policies. I find that the externality increases the welfare effects for the feasible policies and that the minimum volume standard generates larger welfare increases than either of the other feasible policies. Even the minimum volume standard, however, achieves only 9% of the gains from the first-best policy.

More work is still needed in this area to better understand the mechanisms underlying the benefits of market size and their policy implications. For example, research examining how physicians make decisions on where to locate and how to specialize is crucial to better evaluate the welfare consequences of these policies. Further research on other mechanisms that may generate benefits of market size, such as knowledge spillovers, may also provide better understanding of the benefits of large markets and further inform policy.

Despite the future work that is needed, the results in this paper have implications for designing health care policy. The fact that the minimum volume standard generates a much larger welfare gain than the other feasible policies indicates that instituting large price differences across surgeons, such as regulating entry of low-volume surgeons, may be necessary to meaningfully change patient choice and thus welfare. Additionally, the first-best policy results highlight the value of targeting. Since the welfare gain from the first-best policy is substantially larger than that of the coarsely targeted minimum volume standard, setting lower prices for high-quality surgeons, even if imperfectly targeted, may yield even larger gains.

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A Context, data, and measurement appendix

Complication	Measurement period (days from surgery)	% complications
Periprosthetic joint infection/wound infection or other wound complication	90	34.2
Pneumonia or other acute respiratory complication	7	18.9
Pulmonary embolism	30	15.5
Mechanical complication	90	12.6
Death	30	7.1
Acute myocardial infarction	7	6.2
Sepsis/septicemia/shock	7	3.6
Surgical site bleeding or other surgical site complication	30	1.7

Table A1: Complication types for hip and knee replacement

Notes: This table shows each of the eight complication types used to define a complication in my sample, following CMS (2022). The second column shows the period over which this complication is measured, relative to the surgery date. The third column shows the share of total complications that each complication type comprises in my sample.

- Age
- Joint (hip or knee)
- Dually eligible for Medicaid
- Other congenital deformity of hip
- Metastatic cancer and acute leukemia
- Respiratory / heart / digestive / urinary / other neoplasms
- Protein-calorie malnutrition
- Bone / joint / muscle infections / necrosis
- Osteoarthritis of hip or knee
- Dementia or other specified brain disorders
- Hemiplegia, paraplegia, paralysis, functional disability
- Coronary atherosclerosis or angina
- Vascular or circulatory disease
- Pneumonia
- Dialysis status
- Decubitus ulcer or chronic skin ulcer
- Vertebral fractures without spinal cord injury
- Major complications of medical care and trauma
- Gender
- White race
- Multiple procedures
- Post-traumatic osteoarthritis
- Other major cancers
- Diabetes mellitus (DM) or DM complications
- Morbid obesity
- Rheumatoid arthritis and inflammatory connective tissue disease
- Osteoporosis and other bone / cartilage disorders
- Major psychiatric disorders
- Cardio-respiratory failure and shock
- Stroke
- Chronic obstructive pulmonary disease
- Pleural effusion / pneumothorax
- Renal failure
- Trauma
- Other injuries

Table A2: List of patient characteristics and comorbidities

Notes: This table shows all of the patient covariates used to adjust for complications, following CMS (2022). These patient risk covariates are obtained from Medicare claims extending up to twelve months prior to the index admission and include the index admission itself. A binary indicator for whether a patient is white and a binary indicator for if a patient is dually eligible for Medicaid have been added to the covariate list from CMS (2022) specifically for this analysis.

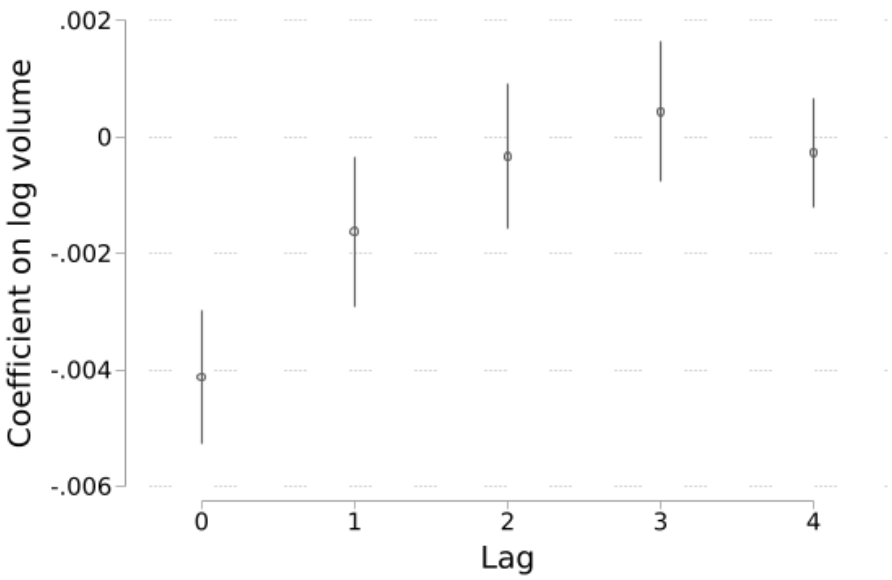


Figure A1: Relationship between probability of a risk-adjusted complication and lags of surgeon hip and knee volume

Notes: This figure shows the relationship between the probability of a risk-adjusted complication and lagged measures of a surgeon's hip and knee volume. Complications are adjusted for year fixed effects and the patient covariates listed in Appendix Table A2. The lags are defined in 365 day intervals. The unit of observation is a patient.

B Descriptive evidence appendix

Outcome	(1) Complication	(2) Complication	(3) Complication	(4) Complication	(5) Complication
Log market size	-0.0023	-0.0031	-0.0033	-0.0025	-0.0013
	(0.0007)	(0.0006)	(0.0007)	(0.0009)	(0.0004)
N	1,978	1,978	1,978	1,972	1,978
Mean complication	0.036	0.035	0.035	0.035	0.012
Risk covariates	X	✓	✓	✓	✓
HRR fixed effects	X	X	✓	X	X
Home market procedure	X	X	X	✓	X
Primary stay complication	X	X	X	X	✓

Table B1: Relationship between complication rate and market size

Notes: This table shows the results from a regression of the complication rate in an HSA on the market size in that HSA averaged over the sample period. The unit of observation is an HSA. Column (1) does not include patient risk covariates. Column (2) adds the risk covariates from Appendix Table A2. Column (3) adds HRR fixed effects. Column (4) only includes patients who receive care in the HSA in which they reside. Column (5) includes only complications that are diagnosed on the same stay-level record as the hip or knee replacement, implying that they occur within the same inpatient stay. Standard errors are clustered at the HSA level.

Outcome	(1) Complication	(2) Complication	(3) Complication	(4) Complication
Log volume $\left(\ln v_{ij}\right)$	-0.0067	-0.0055	-0.0044	-0.0019
	(0.0003)	(0.0003)	(0.0003)	(0.0005)
N	690,798	690,798	690,798	690,798
Mean complication	0.031	0.031	0.031	0.031
Year fixed effects	✓	✓	✓	✓
Risk covariates	X	✓	✓	✓
Hospital fixed effects	X	X	✓	X
Surgeon fixed effects	X	X	X	✓

Table B2: Relationship between probability of a complication and surgeon log hip and knee volume

Notes: This table shows the results from estimating a linear probability model, where the outcome is a binary indicator equal to one if a patient has a complication following a hip or knee replacement and the main independent variable is a surgeon's log hip and knee volume in the 365 days prior to the patient's surgery. Each column includes year fixed effects. Column (2) introduces all the patient covariates shown in Appendix Table A2. Column (3) introduces hospital fixed effects. Column (4) includes surgeon fixed effects but omits hospital fixed effects. The unit of observation is a patient. Standard errors are clustered at the surgeon level.

	(1) Complication	(2) Readmission	(3) Length of stay
Log volume $(\ln v_{ij})$	-0.0055	-0.0049	-0.1897
	(0.0003)	(0.0004)	(0.0093)
N	705,336	705,336	705,336
Outcome mean	0.031	0.051	3.137
Year fixed effects	✓	✓	✓
Risk covariates	✓	✓	✓

Table B3: Relationship between health outcomes and surgeon log hip and knee volume

Notes: This table shows the results from estimating three linear regressions with three different outcomes. The first outcome is a binary indicator equal to one if a patient experiences a complication following a hip or knee replacement. The second outcome is a binary indicator equal to one if a patient is readmitted to a hospital within thirty days following a hip or knee replacement. The third outcome is the patient's length of stay before discharge. In all specifications, the main independent variable is a surgeon's log hip and knee volume in the 365 days prior to the patient's surgery. Each column includes year fixed effects and the patient covariates listed in Appendix Table A2. The unit of observation is a patient. Standard errors are clustered at the surgeon level.

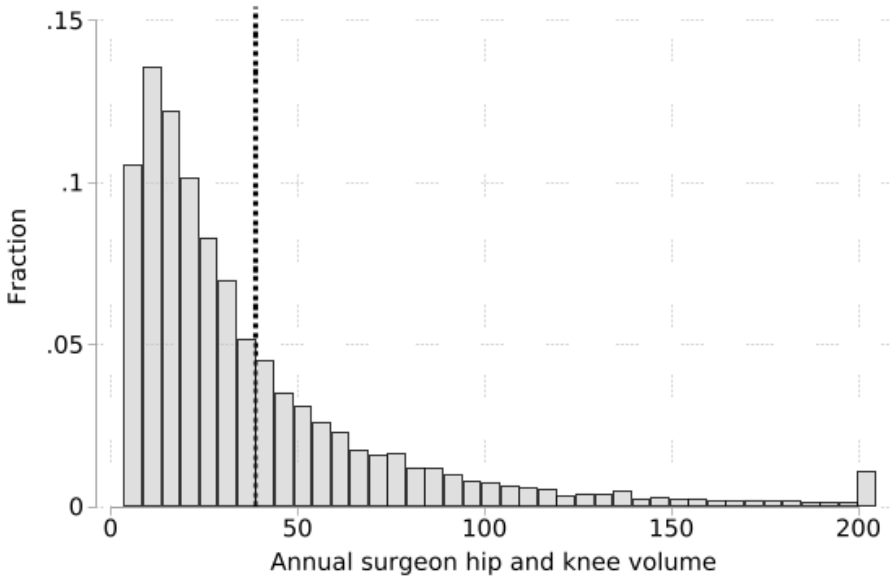


Figure B1: Histogram of surgeon hip and knee volume

Notes: This figure shows the histogram of average surgeon hip and knee volume in the main sample. The unit of observation is a surgeon, and volume is winsorized at the 1% level. The dotted vertical line denotes the median.

C Model and empirical strategy appendix

C.1 Empirical Bayes procedure

Measurement error in the estimated demand and quality parameters poses challenges for both the empirical strategy and the counterfactual policy evaluations, so I use Empirical Bayes shrinkage to address this issue. This procedure reduces the mean-squared error and, crucially for the empirical strategy, eliminates attenuation bias in models using $\widehat{\delta}_{0j}$ as a regressor (Morris, 1983).

To shrink the estimates to the mean, I adjust the parameters according to their standard errors following Morris (1983). For ease of exposition, I discuss the Empirical Bayes shrinkage procedure for estimating δ_{0j}, but the same logic applies to the coefficients, τ and δ_1, as well as to the surgeon fixed effects for complications. Formally, I assume that the estimated fixed effects are estimated with error:

$$\widehat{\delta}_{0j} = \delta_{0j} + e_j,$$

where δ_{0j} is the "true" fixed effect and e_j is the measurement error of the estimated fixed effect. Assume that the errors are independent and drawn from a normal distribution with variance χ_j^2, such that $e_j \sim N\left(0, \chi_j^2\right)$. Hence, the distribution of the estimated fixed effect conditional on the true effect and the measurement error variance is:

$$\widehat{\delta}_{0j}|\delta_{0j}, \chi_j^2 \sim N\left(\delta_{0j}, \chi_j^2\right)$$

Now, if we assume a prior for the true effect and use Bayes' rule, we can derive the shrunk estimate for the parameters as the mean of the posterior. Namely, assume a prior distribution for the true effect such that:

$$\delta_{0j}|\sigma_j^2 \sim N\left(\rho, \sigma^2\right),$$

where ρ is the underlying mean and σ^2 is the variance of the true fixed effect. From Bayes' rule, we have that:

$$\delta_{0j}|\rho, \sigma^2, \chi_j^2, \widehat{\delta}_{0j}\ N\left(b_j\widehat{\delta}_{0j} + (1-b_j)\rho, b_j\chi_j^2\right),$$

where $b_j = \frac{\sigma^2}{\chi_j^2+\sigma^2}$. The Empirical Bayes-adjusted fixed effects correspond to the mean of the posterior, implying that:

$$\delta_{0j}^{EB} = b_j\widehat{\delta}_{0j} + \left(1-b_j\right)\rho.$$

This formulation highlights how the procedure works—the larger the variance of the measurement error, χ_j^2, the more weight is given to the underlying mean, ρ.

Empirically, I use moments from the data and plug them into this formula to shrink these

Empirical Bayes parameters to the mean. From estimating the demand model, I have obtained $\widehat{\delta}_{0j}$, as well as χ_j^2, which is simply the square of the standard error on the fixed effect. I cluster these standard errors at the patient-level to allow for non-independent errors within a patient's different alternatives. I shrink these estimates to the within-market mean, so ρ is simply the mean of the $\widehat{\delta}_{0j}$'s. Finally, by the law of total variance, I calculate the variance of the true fixed effect as $\sigma^2 = Var\left(\widehat{\delta}_{0j}\right) - \mathbb{E}\left[\chi_j^2\right]$, where $Var\left(\widehat{\delta}_{0j}\right)$ is the variance of the fixed effects and $\mathbb{E}\left[\chi_j^2\right]$ is the mean of the individual fixed effect variances. I perform a similar calculation for the surgeon fixed effects for complication rates and then for τ and δ_1 within markets.

C.2 Deriving the control function

Recall the conditional expectation of the estimating equation error term, ϵ_{ij} from Equation 3 is:

$$\mathbb{E}\left[\epsilon_{ij}|v_{ij}, X_i, \gamma_{t(i)}, \widehat{\delta}_{0j}, \eta_{i1}, \cdots, \eta_{iJ}, D_i = j\right] = \kappa\widehat{\delta}_{0j} + \sum_k \phi_k \tilde{\eta}_{ik} + \varphi\tilde{\eta}_{ij},$$

where v_{ij} is surgeon j's volume (plus one) in the 365 days prior to patient i's surgery date, X_i are the patient risk covariates listed in Appendix Table A2, $\gamma_{t(i)}$ are year fixed effects, $\widehat{\delta}_{0j}$ is the estimated demand for exogenous surgeon quality from the demand model in Equation 1, $\tilde{\eta}_{ij} = \eta_{ij} - \mu_\eta$ are the mean-zero logit shocks from the demand model, J is the total number of surgeons, and D_i denotes patient i's chosen surgeon. We want to show that the expected values of the unobserved logit shocks, $\eta_{i1}, \cdots, \eta_{iJ}$ conditional on the choice, yields the following conditional expectation function:

$$\mathbb{E}\left[\epsilon_{ij}|v_{ij}, X_i, \gamma_{t(i)}, \widehat{\delta}_{0j}, d_i, D_i = j\right] = \kappa\widehat{\delta}_{0j} + \sum_k \phi_k \theta_{ik}\left(j, d_i\right) + \varphi\theta_{ij}\left(j, d_i\right),$$

where $d_i = (d_{i1}, \cdots, d_{iJ})$ is the vector of distances between patient i and each surgeon that functions as the excluded instrument and $\theta_{ik}(j, d_i)$ are simply functions of the logit choice probabilities from the demand model:

$$\theta_{ik}(j) = \begin{cases} -\ln \widehat{p}_{ik}\left(d_i\right), & k = j \\ \frac{\widehat{p}_{ik}(d_i)}{1-\widehat{p}_{ik}(d_i)} \ln \widehat{p}_{ik}\left(d_i\right), & \text{otherwise} \end{cases},$$

Here, $\widehat{p}_{ij}(d_i)$ is the predicted probability that patient i chooses surgeon j as a function of the differential distance instrument. Because $\widehat{\delta}_{0j}$ does not depend on the logit shocks, I focus on deriving the latter two terms. I derive the cases when $k = j$ and when $k \neq j$ separately.

I first begin this derivation by showing two crucial relationships. First, note that:

$$\mathbb{E}\left[\tilde{\eta}_{ij}|D_i = j\right] = \mathbb{E}\left[u_{ij}|D_i = j\right] - \delta_{0j} - \delta_1 \ln v_{ij} + \tau \ln d_{ij} - \mu_\eta \tag{C1}$$

Additionally, using Small and Rosen (1981), we have:

$$\mathbb{E}\left[u_{ij}|D_i = j\right] = \ln\left[\sum_{s=1}^{J} \exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right)\right] + \mu_\eta \tag{C2}$$

Now, we can solve for the control function for the chosen alternative when $k = j$. Substituting

Equation C2 into Equation C1 yields:

$$\begin{aligned}
\mathbb{E}\left[\tilde{\eta}_{ij}|D_i = j\right] &= \ln\left[\sum_{s=1}^{J}\exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right)\right] + \mu_\eta - \delta_{0j} - \delta_1 \ln v_{ij} + \tau \ln d_{ij} - \mu_\eta \\
&= \ln\left[\sum_{s=1}^{J}\exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right)\right] - \delta_{0j} - \delta_1 \ln v_{ij} + \tau \ln d_{ij} \\
&= \ln\left[\sum_{s=1}^{J}\exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right)\right] - \ln\left[\exp\left(\delta_{0j} + \delta_1 \ln v_{ij} - \tau \ln d_{ij}\right)\right] \\
&= \ln\left[\frac{\sum_{s=1}^{J}\exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right)}{\exp\left(\delta_{0j} + \delta_1 \ln v_{ij} - \tau \ln d_{ij}\right)}\right] \\
&= -\ln\left[\frac{\exp\left(\delta_{0j} + \delta_1 \ln v_{ij} - \tau \ln d_{ij}\right)}{\sum_{s=1}^{J}\exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right)}\right] \\
&= -\ln \widehat{p}_{ij}\left(d_i\right)
\end{aligned}$$

We can also solve for the control function for a non-chosen alternative, such that $k \neq j$. Re-writing Equation C1 for a non-chosen alternative yields:

$$\mathbb{E}\left[\tilde{\eta}_{ik}|D_i = j\right] = \mathbb{E}\left[u_{ik}|D_i = j\right] - \delta_{0k} - \delta_1 \ln v_{ik} + \tau \ln d_{ik} - \mu_\eta \tag{C3}$$

Additionally, by the properties of conditional expectation, we have that:

$$\mathbb{E}\left[u_{ik}\right] = \mathbb{P}\left(D_i = k\right)\mathbb{E}\left[u_{ik}|D_i = k\right] + \mathbb{P}\left(D_i \neq k\right)\mathbb{E}\left[u_{ik}|D_i \neq k\right]$$

Then, substituting the relationship in Equation C2 into this relationship yields:

$$\begin{aligned}
&\delta_{0k} + \delta_1 \ln v_{ik} - \tau \ln d_{ik} + \mu_\eta = \\
&\mathbb{P}\left(D_i = k\right)\left[\ln\left[\sum_{s=1}^{J}\exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right)\right] + \mu_\eta\right] + \mathbb{P}\left(D_i \neq k\right)\mathbb{E}\left[u_{ik}|D_i \neq k\right].
\end{aligned}$$

Therefore, we have that:

$$\begin{aligned}
\mathbb{E}\left[u_{ik}|D_i \neq k\right] = \frac{1}{1 - \mathbb{P}\left(D_i = k\right)}\Bigg(&\delta_{0k} + \delta_1 \ln v_{ik} - \tau \ln d_{ik} + \mu_\eta \\
&- \mathbb{P}\left(D_i = k\right)\left[\ln\left[\sum_{s=1}^{J}\exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right)\right] + \mu_\eta\right]\Bigg)
\end{aligned} \tag{C4}$$

We can then substitute in these two expressions to solve for the control function term for a non-chosen alternative. Specifically, from Equation C4, we know the conditional expectation of u_{ik} conditional on the choice not being k. Supposing that the choice is j, we can then plug Equation

C4 into Equation C3:

$$
\begin{aligned}
\mathbb{E}\left[\tilde{\eta}_{ik} | D_i = j\right] = & \frac{1}{1 - \mathbb{P}\left(D_i = k\right)} \Bigg(\delta_{0k} + \delta_1 \ln v_{ik} - \tau \ln d_{ik} + \mu_\eta \\
& - \mathbb{P}\left(D_i = k\right) \left[\ln \left[\sum_{s=1}^{J} \exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right) \right] + \mu_\eta \right] \Bigg) - \delta_{0k} - \delta_1 \ln v_{ik} + \tau \ln d_{ik} - \mu_\eta \\
= & \frac{1}{1 - \mathbb{P}\left(D_i = k\right)} \Bigg(\delta_{0k} + \delta_1 \ln v_{ik} - \tau \ln d_{ik} + \mu_\eta - \left(1 - \mathbb{P}\left(D_i = k\right)\right) \left[\delta_{0k} + \delta_1 \ln v_{ik} - \tau \ln d_{ik} + \mu_\eta \right] \\
& - \mathbb{P}\left(D_i = k\right) \left[\ln \left[\sum_{s=1}^{J} \exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right) \right] + \mu_\eta \right] \Bigg) \\
= & \frac{\mathbb{P}\left(D_i = k\right)}{1 - \mathbb{P}\left(D_i = k\right)} \left(\delta_{0k} + \delta_1 \ln v_{ik} - \tau \ln d_{ik} - \ln \left[\sum_{s=1}^{J} \exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right) \right] \right) \\
= & \frac{\mathbb{P}\left(D_i = k\right)}{1 - \mathbb{P}\left(D_i = k\right)} \left(\ln \left[\exp\left(\delta_{0k} + \delta_1 \ln v_{ik} - \tau \ln d_{ik}\right) \right] - \ln \left[\sum_{s=1}^{J} \exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right) \right] \right) \\
= & \frac{\mathbb{P}\left(D_i = k\right)}{1 - \mathbb{P}\left(D_i = k\right)} \left(\ln \frac{\exp\left(\delta_{0k} + \delta_1 \ln v_{ik} - \tau \ln d_{ik}\right)}{\sum_{s=1}^{J} \exp\left(\delta_{0s} + \delta_1 \ln v_{is} - \tau \ln d_{is}\right)} \right) \\
= & \frac{\widehat{p}_{ik}\left(d_i\right)}{1 - \widehat{p}_{ik}\left(d_i\right)} \ln \widehat{p}_{ik}\left(d_i\right),
\end{aligned}
$$

where $\mathbb{P}\left(D_i = k\right)\left(d_i\right) = \widehat{p}_{ik}(d_i)$ by definition.

C.3 Estimation procedure

Directly estimating the main empirical specification in Equation 7 is computationally burdensome, so I leverage the Frisch-Waugh-Lovell theorem to estimate the parameters. In particular, because Equation 7 has a parameter for each surgeon in the dataset, of which there are more than 11,000, I cannot estimate it quickly via OLS. I therefore exploit the fact that the θ_{ik}'s are zero for surgeons outside a patient's HRR and use the Frisch-Waugh-Lovell theorem to obtain unbiased estimates of the parameters. Namely, I partition the parameters into the local parameters, which are the θ_{ik}'s, and the national parameters, which are all the other parameters. I first residualize out the local parameters from the national parameters within a market. I then aggregate these residualized national parameters to run one national-level regression and obtain coefficients on all national parameters. To recover the local parameters, I estimate the full estimating equation at the market-level holding fixed the national parameters at their estimated values.

C.4 Correcting for measurement error

Because there is measurement error in the key explanatory variable, volume, the estimate of the returns to surgeon volume may be biased. Measurement error comes from two sources. The first is classical measurement error and reflects the fact that I only observe a 20% random sample of Medicare FFS volume. The second may be classical or non-classical and reflects the fact that I only observe Medicare FFS volume, not all-payer volume, including Medicare Advantage volume. Using properties of random sampling and a validation dataset from the state of Florida, I show that this measurement error attenuates the estimated returns to surgeon volume parameter, such that the parameter I estimate is only 67% of the true value.

C.4.1 Measurement error due to random sampling

Because I only observe a random 20% sampling of Medicare FFS claims, the estimated returns to surgeon volume parameter is attenuated. Since this measurement error is approximately classical, I compute a correction factor to purge the estimate of this attenuation using a classical errors-in-variables model and the properties of random sampling.

Using a classic errors-in-variables formula, I compute the correction factor needed to adjust for the attenuation bias in measured volume due to random sampling. Let $v_{obs} = \ln(N_{obs} + 1)$ denote the log Medicare FFS volume (plus one) that I observe, $v_{samp} = \ln(pN_{FFS} + 1)$ denote the log of the expected Medicare FFS volume (plus one) in the random sample given some sampling probability p, and e denote the measurement error from the random sampling. Following the classical errors-in-variables model, I assume that the error enters additively (in logs), implying $v_{obs} = v_{samp} + e$. Then, the correction factor, or reliability ratio, from random sampling c_{samp} is given by:

$$c_{samp} = \frac{\sigma^2_{v_{samp}}}{\sigma^2_{v_{samp}} + \sigma^2_e}$$

where $\sigma^2_{v_{samp}}$ is the variance of v_{samp} and σ^2_e is the variance of e. This correction factor is the amount by which the estimate is attenuated and thus can be used to recover the unattenuated estimate.

Using the properties of the measurement error, I reformulate the correction factor into an object I can actually compute. The issue is that while c_{samp} would correct for the attenuation bias, v_{samp} is not observed, so I cannot compute $\sigma^2_{v_{samp}}$. However, because the error term enters additively in logs and since the covariance between the true variable and the measurement error is approximately zero due to random sampling, the variance of the measured variable, v_{obs}, is just the sum of the variance of the true variable v_{samp} and the variance of the error term, e: $\sigma^2_{v_{obs}} =$

$\sigma^2_{v_{samp}} + \sigma^2_e$. Hence, the correction factor due to random sampling, c_{samp} can be re-expressed as:

$$c_{samp} = 1 - \frac{\sigma^2_e}{\sigma^2_{v_{obs}}}.$$

I compute the variance components of this correction factor using properties of the random sampling and the data and show that the returns to surgeon volume is 79% of the true value due to random sampling. Computing $\sigma^2_{v_{obs}}$ is simple, as it is just the variance of the residualized log-volume. To compute σ^2_e, note that by the law of total variance $\sigma^2_e = \mathbb{E}\left[Var\left[e|v_{samp}\right]\right] + Var\left[\mathbb{E}\left[e|v_{samp}\right]\right] = \mathbb{E}\left[Var\left[e|v_{samp}\right]\right]$, since $\mathbb{E}\left[e|v_{samp}\right] \approx 0$. Then, using the definition of the classical measurement error, note that $Var\left[e|v_{samp}\right] = Var\left[v_{obs} - v_{samp}|v_{samp}\right] = Var\left[v_{obs}|v_{samp}\right]$. Next, because the sample I use is a 20% random sample, I can model the distribution of N conditional on N_{FFS} as a binomial distribution with probability of success $p = 0.20$, which implies that $N_{obs}|N_{FFS} \sim binomial(N_{FFS}, p)$. Recalling that v_{obs} is the log of volume plus one, we can use the known variance of a binomial distribution and the delta method to calculate the expected value of this variance:

$$\begin{aligned}
\mathbb{E}\left[Var\left[v_{obs}|v_{samp}\right]\right] &= \mathbb{E}\left[Var\left[\ln(1+N_{obs})|N_{FFS}\right]\right] \\
&= \mathbb{E}\left[\left(\frac{1}{pN_{FFS}+1}\right)^2 \times \left[pN_{FFS} \times (1-p)\right]\right] \\
&= \mathbb{E}\left[\left(\frac{1}{\frac{pN_{obs}}{p}+1}\right)^2 \times \left[\frac{pN_{obs}}{p} \times (1-p)\right]\right] \\
&= \mathbb{E}\left[\left(\frac{1}{N_{obs}+1}\right)^2 \times \left[N_{obs} \times (1-p)\right]\right].
\end{aligned}$$

The second equality follows from both using the delta method where $f(x) = \ln(x+1)$ is expanded around $E[N_{obs}|N_{FFS}] = pN_{FFS}$ and the known variance of a binomial distribution. Meanwhile, the third equality follows from approximating N_{FFS} with $\frac{N_{obs}}{p}$.[36] Computing this final expression is relatively easy, since I can just take the mean of this function of the observed volume in my sample. Using this formula to calculate σ^2_e, combined with the calculation of $\sigma^2_{v_{obs}}$, implies that $c_{samp} = 0.79$, such that the returns to surgeon volume estimate is 79% of the true value due to attenuation bias from random sampling in my main sample.

C.4.2 Measurement error due to payer sampling

Because I only observe a surgeon's Medicare FFS volume and not their other payer volume, there may be additional bias in the estimated returns to surgeon volume parameter. Since this bias may be classical or non-classical, I solve for the resulting bias allowing for correlations between the

[36] Because N_{obs} is a noisy measure of pN_{FFS}, this approximation understates the sampling variance for low-volume surgeons. Thus, the correction factor here is conservatively large.

measurement error and the true value using a validation dataset from the state of Florida.

To calculate the correction factor from this measurement error, I introduce and interpret the formula for non-classical measurement error. Now, let $v_{FFS} = \ln(N_{FFS} + 1)$ denote a surgeon's Medicare FFS log volume (plus one). Note, that I treat v_{FFS} as observed, since it is in the validation dataset. Additionally, let v^* denote log all-payer volume (plus one), and let u denote the measurement error from only observing Medicare FFS as a payer. For non-classical measurement error, we have that the correction factor due to only observing Medicare as a payer, c_{payer} is:

$$c_{payer} = \frac{\sigma^2_{v^*} + \sigma_{v^*u}}{\sigma^2_{v_{FFS}}},$$

where $\sigma^2_{v^*}$ is the variance of the true volume, σ_{v^*u} is the covariance between the true volume and the measurement error, and $\sigma^2_{v_{FFS}}$ is the variance of the Medicare FFS volume. This formula is simply an alternate version of the omitted variable bias, where non-Medicare FFS volume is the omitted variable that may potentially be correlated with levels of the Medicare FFS volume. Thus, another way to compute this correction factor is by regressing the error, u, on the Medicare FFS volume, v_{FFS}. This formula also nests the formula with classical measurement error in which $\sigma_{v^*u} = 0$. Thus, the magnitude of σ_{v^*u} governs the influence of non-classical measurement error on the bias.

I compute the correction term due to only observing Medicare FFS as a payer using a validation dataset from the state of Florida. Each of these objects is straightforward to calculate in the data. While this validation exercise provides significant benefits, there are two important caveats. First, because I do not observe all the controls in this validation dataset, I cannot residualize them out, so I assume the unconditional moments are good proxies for the conditional moments. Second, I assume that the relationship between Medicare FFS and all-payer volume in Florida generalizes to other states. The final calculation yields $c_{payer} = 0.85$, indicating that the returns to surgeon volume estimate is 85% of the true value due to bias from only measuring Medicare FFS volume. Had I assumed the measurement error was instead classical and omitted the σ_{v^*u} term, c_{payer} would be 0.87, indicating that the non-classical measurement error introduces slightly more attenuation, as the measurement error is larger for higher (all-payer) volume surgeons. However, the magnitude of this non-classical measurement error is quite small, as evidenced in the small difference between the two correction factors with and without σ_{v^*u}.

C.4.3 Combining both measurement errors together

The final correction coefficient implies that the estimated parameter is 67% of the true value, yet in practice, I only correct for the measurement error due to random sampling. Since the measurement error from random sampling and from only observing Medicare FFS volume are independent, the bias is multiplicative in the correction terms. That is, the final correction term is $c = c_{samp} \times c_{payer} = 0.67$, implying that the estimated returns to surgeon volume parameter is 67% of the underlying structural parameter. To be consistent with the descriptive magnitudes I use for the quantification

exercises and to avoid relying on the stronger assumptions from using the validation dataset, in practice, I only use the conversion coefficient from the random sampling. Additionally, to be consistent across specifications, I use this measurement error correction for all specifications, even if there are small differences in the residualized variance of log volume.

C.5 Calculating standard errors

Calculating standard errors in this setting is difficult since the first stage is a logit model that is computationally burdensome to estimate. To address this challenge, I use a two-step score bootstrap procedure originally developed in Kline and Santos (2012) and used in other similar settings (Abdulkadiroğlu et al., 2020; Einav, Finkelstein, and Mahoney, 2022). This procedure adjusts inference for the extra uncertainty introduced by the demand model estimates without recalculating the first-stage estimates or analytically deriving the influence of the first-stage estimates on the second-stage estimates. Intuitively, it yields perturbed values of the estimates by randomly weighting observations.

I first introduce simplified notation to help ease the exposition of this procedure. Using the demand model in Equation 1, I define Δ as the vector of market-specific parameters estimated via maximum likelihood from this first stage:

$$\Delta = \left(\delta_{0j}, \cdots, \delta_{0|S_i|}, \delta_1, \tau\right),$$

where $|S_i|$ denotes the cardinality of the set of surgeons within a market. Similarly, for the second stage, define Γ as the vector of parameters from Equation 7:

$$\Gamma = \left(\beta_0, \beta_1, \alpha, \gamma_{t(i)}, \kappa, \phi_1, \cdots, \phi_{|S_i|}, \varphi, \mu_m\right)$$

To generate bootstrap values of these parameters, I first perturb the first-stage estimates using random weights. Specifically, I generate a bootstrap distribution by taking repeated Newton-Raphson steps from the full-sample estimates and randomly reweighting each observation's score contribution. To cluster the standard errors at the surgeon level, I draw the same weights for all patients who choose the same surgeon. The first-stage bootstrap estimate of Δ in bootstrap replication b is:

$$\widehat{\Delta}^b = \widehat{\Delta} - H_1^{-1}\left(\widehat{\Delta}\right) \times \sum_{i=1}^{N} w_i^b s_1\left(i, \widehat{\Delta}\right),$$

where $H_1\left(\widehat{\Delta}\right)$ is the first-stage Hessian, N is the number of patients, $s_1\left(i, \widehat{\Delta}\right)$ is the first-stage score contribution of patient i at the optimal parameter vector $\widehat{\Delta}$, and $w_i^b = w_{j(i)}^b$ and $w_{j(i)}^b \sim N(0,1)$ are i.i.d. standard normal weights for patient i's chosen surgeon j.

Then, using the same random weights and the perturbed first-stage values, I generate perturbed values for the second-stage bootstrap estimates. Namely, using another set of Newton-Raphson steps, I define a second-stage bootstrap estimate Γ in bootstrap replication b as:

$$\widehat{\Gamma}^b = \widehat{\Gamma} - H_2^{-1}\left(\widehat{\Gamma}; \widehat{\Delta}\right) \times \sum_{i=1}^{N} \left[w_i^b s_2\left(i, \widehat{\Gamma}; \widehat{\Delta}\right) - s_2\left(i, \widehat{\Gamma}; \widehat{\Delta}^b\right)\right], \tag{C5}$$

where $H_2\left(\widehat{\Gamma};\widehat{\Delta}\right)$ is the second-stage Hessian and $s_2\left(i,\widehat{\Gamma};\widehat{\Delta}\right)$ is the second-stage score contribution of patient i at the two-stage optima $\left(\widehat{\Gamma};\widehat{\Delta}\right)$. Intuitively, the term $s_2\left(i,\widehat{\Gamma};\widehat{\Delta}^b\right)$ accounts for the additional variability in the second-stage score due to the first-stage estimate.

To estimate these objects, I use standard econometric identities to exploit the fact that most of these objects are already estimated. Namely, I compute $H_1^{-1}\left(\widehat{\Delta}\right)$ as the variance-covariance matrix divided by the residual variance from the first-stage logit estimation for a given market. Meanwhile, I compute the first-stage score contribution as the value of the covariate multiplied by the residual. For the second stage, I compute $H_2^{-1}\left(\widehat{\Gamma};\widehat{\Delta}\right)$ as the variance-covariance matrix after partialling out the market-specific parameters, as detailed in my estimation strategy in Appendix C.3. Note that this result relies on the equivalence of cluster-robust standard errors between the partialled-out regression and the full regression (Ding, 2021). Finally, I construct the score vectors from the second stage by multiplying the residuals from the partialled out regression by the value of either the covariate (for $s_2\left(i,\widehat{\Gamma};\widehat{\Delta}\right)$) or the perturbed covariate (for $s_2\left(i,\widehat{\Gamma};\widehat{\Delta}^b\right)$). Finally, to account for the adjustment due to measurement error described in Appendix C.4, I multiply the standard errors from the two-step score bootstrap procedure by the measurement error correction term, which treats this term as a constant for computational tractability.

C.6 Constructing an amenity index

To address the potential threat from the endogeneity of surgeon locations, I control for various characteristics of larger markets that may disproportionately attract higher exogenous quality surgeons. One of these controls is the amenity value of the ZIP code in which the surgeon practices. To construct this amenity index, I use data at the Census block group-level from the EPA Smart Location Database in 2013. In particular, I use nine variables that are related to density and perhaps relevant for a surgeon's location decision: residential density, the percent of zero-car households, the share of high-wage earners (earning more than $3,333 per month), the share of retail jobs, the share of service jobs, the share of entertainment jobs, total road network density, network density in terms of facility miles of pedestrian-oriented links per square mile, and the proportion of employment within one-quarter of a mile of a fixed-guideway transit stop. I then aggregate this data to the ZIP code-level based on the share of the block group's 2010 population that lives in that ZIP code. Using these nine different variables, I conduct a Principal Component Analysis and extract the first component. This first component is the amenity index that I use in the analysis.

C.7 Tables

	(1) $\widehat{p}_{ij}$	(2) $\widehat{p}_{ij}$	(3) $\widehat{p}_{ij}$
Predicted probability ($\widehat{p}_{ij}$)	0.9059	0.9534	0.8922
	(0.0041)	(0.0029)	(0.0035)
N	39,243	39,243	39,243
Distance functional form	Square	Square root	Quintiles

Table C1: Relationship between predicted probabilities with alternative demand model functional forms for distance in Springfield, Massachusetts

Notes: This table shows the relationship between the predicted choice probabilities, $\widehat{p}_{ij}$, in the demand model from Equation 1 and the predicted choice probabilities under alternative functional form specifications for distance in the demand model for the Springfield, Massachusetts HRR. The outcome in these regressions is $\widehat{p}_{ij}$ under the main demand model, while the independent variables are the $\widehat{p}_{ij}$'s from the alternative functional forms. In column (1), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} - \tau_1 d_{ij} + \tau_2 d_{ij}^2 + \eta_{ij}$. In column (2), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} - \tau \sqrt{d_{ij}} + \eta_{ij}$. In column (3), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} - \tau_1 d_{ij}^{\text{bin 2}} - \tau_2 d_{ij}^{\text{bin 3}} - \tau_3 d_{ij}^{\text{bin 4}} - \tau_4 d_{ij}^{\text{bin 5}} + \eta_{ij}$, where $d_{ij}^{\text{bin k}}$ denotes the kth quintile of distance and the first quintile is the reference group. Standard errors are clustered at the patient level.

	(1) $\widehat{\delta}_{0j}$	(2) $\widehat{\delta}_{0j}$	(3) $\widehat{\delta}_{0j}$
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$	1.1497	1.0483	1.1324
	(0.0342)	(0.0135)	(0.0489)
N	33	33	33
Distance functional form	Square	Square root	Quintiles

Table C2: Relationship between demand for exogenous surgeon quality with alternative demand model functional forms for distance in Springfield, Massachusetts

Notes: This table shows the relationship between the demand for exogenous surgeon quality, $\widehat{\delta}_{0j}$, in the demand model from Equation 1 and the demand for exogenous surgeon quality under alternative functional form specifications for distance in the demand model for the Springfield, Massachusetts HRR. The outcome in these regressions is $\widehat{\delta}_{0j}$ under the main demand model, while the independent variables are the $\widehat{\delta}_{0j}$'s from the alternative functional forms. In column (1), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} - \tau_1 d_{ij} + \tau_2 d_{ij}^2 + \eta_{ij}$. In column (2), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} - \tau \sqrt{d_{ij}} + \eta_{ij}$. In column (3), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} - \tau_1 d_{ij}^{\text{bin 2}} - \tau_2 d_{ij}^{\text{bin 3}} - \tau_3 d_{ij}^{\text{bin 4}} - \tau_4 d_{ij}^{\text{bin 5}} + \eta_{ij}$, where $d_{ij}^{\text{bin k}}$ denotes the kth quintile of distance and the first quintile is the reference group. Standard errors are clustered at the patient level.

C.8 Figures

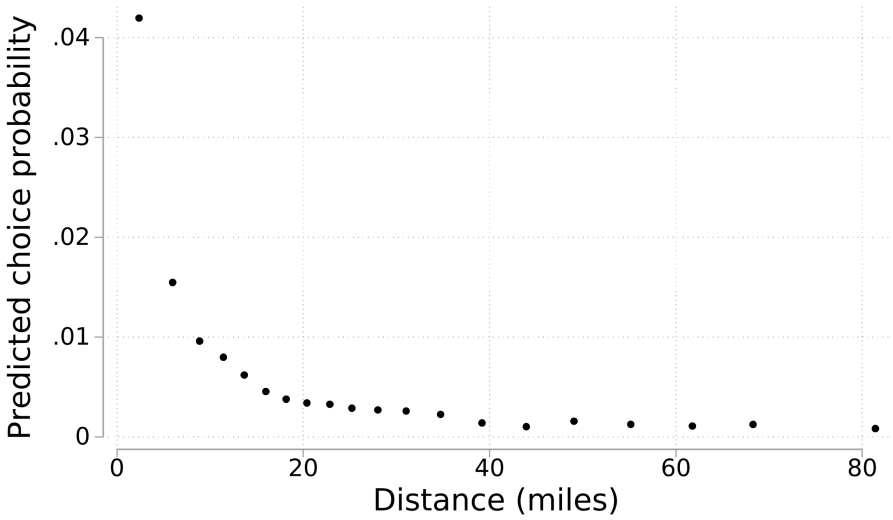


Figure C1: Relationship between predicted probability of choosing surgeon and distance in Boston HRR

Notes: This figure shows the relationship between the predicted probability of choosing a surgeon as estimated from Equation 1 and the distance to the surgeon in the Boston HRR. The unit of observation is a patient-surgeon.

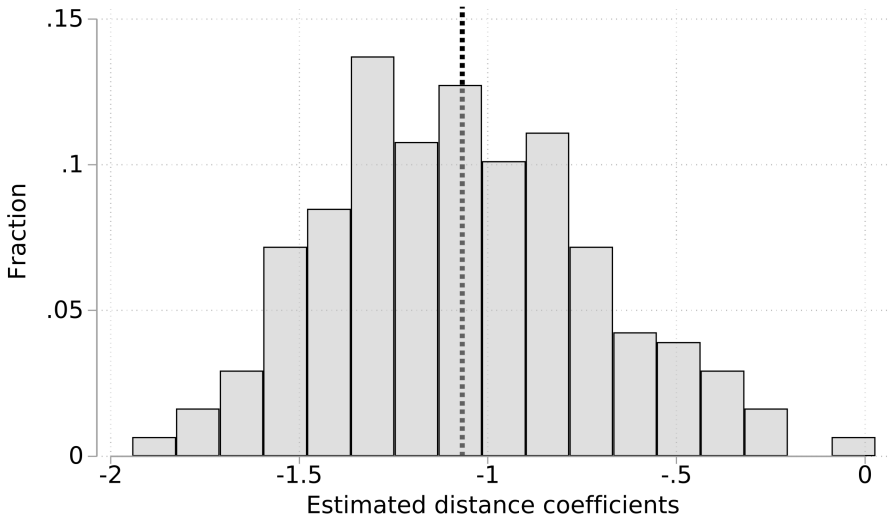


Figure C2: Histogram of estimated distance coefficients across markets

Notes: This figure shows the histogram of estimated distance coefficients for each HRR, as estimated from Equation 1. The coefficients have been shrunk to the national mean using the Empirical Bayes procedure described in Appendix Section C.1. The dotted vertical line denotes the mean.

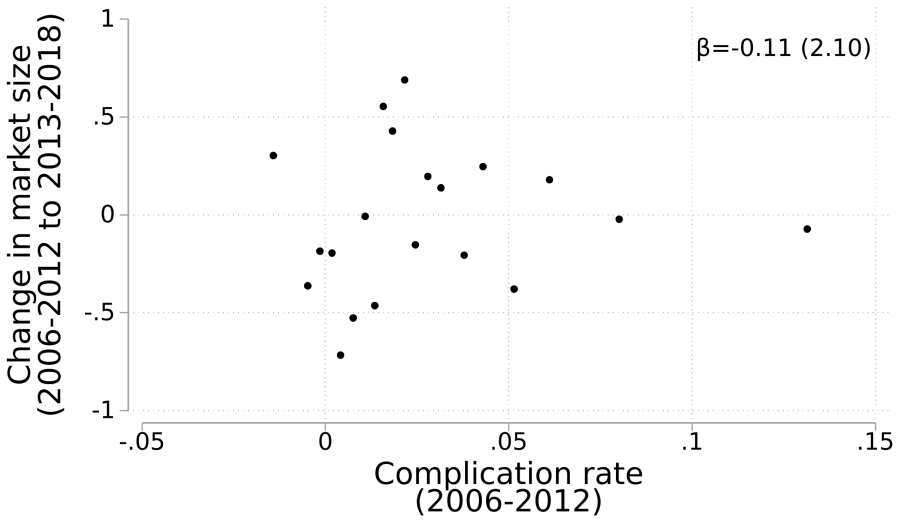


Figure C3: Relationship between change in HSA market size and surgeon's complication rate

Notes: This figure shows the relationship between the change in market size of a surgeon's primary practice HSA from 2006–2012 to 2013–2018 versus the surgeon's risk-adjusted complication rate from 2006–2012 for 521 surgeons who move HSAs within the same HRR. A surgeon's primary practice HSA is the HSA where the surgeon performs a plurality of hip and knee replacements over the time period. Market size is the average market size of the HSA over the whole time period. A surgeon's risk-adjusted complication is a surgeon fixed effect from a linear probability model, where the outcome is a binary indicator equal to one if a patient experiences a complication following the hip or knee replacement and the independent variables are the covariates shown in Appendix Table A2, as well as year fixed effects. The unit of observation is a surgeon.

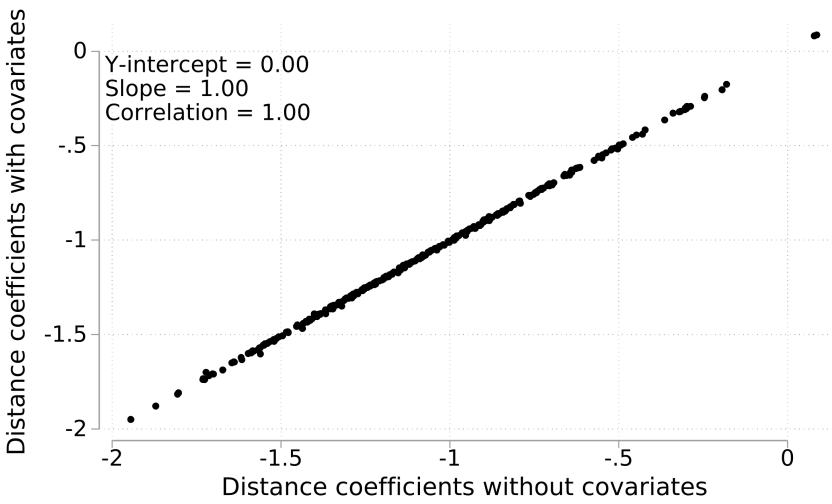


Figure C4: Estimated distance coefficient with vs. without patient covariates for HRRs

Notes: This figure shows the relationship between the estimated distance coefficients with and without patient covariates for the 306 HRRs. The estimated distance coefficient, τ, from the demand model in Equation 1 is plotted on the x-axis. The estimated distance coefficient, τ, from a demand model in which $\delta_{jt}(v_{jt}) = \delta_{0j} + \delta_{1j}\widehat{X}_{it} + \delta_1 \ln v_{jt}$ in Equation 1 is plotted on the y-axis. The composite risk index, X_i, is the first component of a Principal Component Analysis with all the risk covariates listed in Appendix Table A2.

D Results: The returns to surgeon volume and the benefits of market size appendix

	(1) $\widehat{p}_{ij}$	(2) $\widehat{p}_{ij}$	(3) $\widehat{p}_{ij}$	(4) $\widehat{p}_{ij}$	(5) $\widehat{p}_{ij}$
Predicted probability ($\widehat{p}_{ij}$)	0.9904	1.0002	0.9995	0.9970	0.9999
	(0.0007)	(0.0001)	(0.0006)	(0.0005)	(0.0001)
N	37,317	35,664	39,243	39,243	39,243
Volume functional form	Lag	No zeros	Quintiles	Hip and knee	Non-hip and knee

Table D1: Relationship between predicted probabilities with alternative demand model functional forms for volume in Springfield, Massachusetts

Notes: This table shows the relationship between the predicted choice probabilities, $\widehat{p}_{ij}$, in the demand model from Equation 1 and the predicted choice probabilities under alternative functional form specifications for volume in the demand model for the Springfield, Massachusetts HRR. The outcome in these regressions is $\widehat{p}_{ij}$ under the main demand model, while the independent variables are the $\widehat{p}_{ij}$'s from the alternative functional forms. In column (1), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} + \delta_2 \ln v_{ij}^{lag} - \tau \ln d_{ij} + \eta_{ij}$, where v_{ij}^{lag} is a surgeon's lagged log hip and knee volume (the surgeon's hip and knee volume from 730 days prior to the surgery date to 365 days prior). In column (2), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} - \tau \ln d_{ij} + \eta_{ij}$, where v_{ij} is no longer one plus hip and knee volume but just hip and knee volume (for the outside option, utility is still normalized to zero). In column (3), the demand model is $u_{ij} = \delta_{0j} + \delta_1 v_{ij}^{\text{bin } 2} + \delta_2 v_{ij}^{\text{bin } 3} + \delta_3 v_{ij}^{\text{bin } 4} + \delta_4 v_{ij}^{\text{bin } 5} - \tau \ln d_{ij} + \eta_{ij}$, where v_{ij}^{k} denotes the kth quintile of a surgeon's hip and knee volume and the first quintile is the reference group. In column (4), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij}^{hip} + \delta_2 \ln v_{ij}^{knee} - \tau \ln d_{ij} + \eta_{ij}$, where v_{ij}^{hip} and v_{ij}^{knee} are surgeon j's hip and knee volume plus one, respectively, in the 365 days prior to patient i's surgery date. In column (5), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} + \delta_2 \ln v_{ij}^{revision} + \delta_3 \ln v_{ij}^{arthroscopy} - \tau \ln d_{ij} + \eta_{ij}$, where $v_{ij}^{revision}$ and $v_{ij}^{arthroscopy}$ denote surgeon j's hip and knee revision and knee arthroscopy volume plus one, respectively. Standard errors are clustered at the patient level.

	(1) $\widehat{\delta}_{0j}$	(2) $\widehat{\delta}_{0j}$	(3) $\widehat{\delta}_{0j}$	(4) $\widehat{\delta}_{0j}$	(5) $\widehat{\delta}_{0j}$
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$	0.8684	1.0482	0.9251	0.9816	1.0123
	(0.0054)	(0.0036)	(0.0145)	(0.0168)	(0.0032)
N	33	33	33	33	33
Volume functional form	Lag	No zeros	Quintiles	Hip and knee	Non-hip and knee

Table D2: Relationship between demand for exogenous surgeon quality with alternative demand model functional forms for volume in Springfield, Massachusetts

Notes: This table shows the relationship between the demand for exogenous surgeon quality, $\widehat{\delta}_{0j}$, in the demand model from Equation 1 and the demand for exogenous surgeon quality under alternative functional form specifications for volume in the demand model for the Springfield, Massachusetts HRR. The outcome in these regressions is $\widehat{\delta}_{0j}$ under the main demand model, while the independent variables are the $\widehat{\delta}_{0j}$'s from the alternative functional forms. In column (1), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} + \delta_2 \ln v_{ij}^{lag} - \tau \ln d_{ij} + \eta_{ij}$, where v_{ij}^{lag} is a surgeon's lagged log hip and knee volume (the surgeon's hip and knee volume from 730 days prior to the surgery date to 365 days prior). In column (2), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} - \tau \ln d_{ij} + \eta_{ij}$, where v_{ij} is no longer one plus hip and knee volume but just hip and knee volume (for the outside option, utility is still normalized to zero). In column (3), the demand model is $u_{ij} = \delta_{0j} + \delta_1 v_{ij}^{\text{bin } 2} + \delta_2 v_{ij}^{\text{bin } 3} + \delta_3 v_{ij}^{\text{bin } 4} + \delta_4 v_{ij}^{\text{bin } 5} - \tau \ln d_{ij} + \eta_{ij}$, where v_{ij}^{k} denotes the kth quintile of a surgeon's hip and knee volume and the first quintile is the reference group. In column (4), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij}^{hip} + \delta_2 \ln v_{ij}^{knee} - \tau \ln d_{ij} + \eta_{ij}$, where v_{ij}^{hip} and v_{ij}^{knee} are surgeon j's hip and knee volume plus one, respectively, in the 365 days prior to patient i's surgery date. In column (5), the demand model is $u_{ij} = \delta_{0j} + \delta_1 \ln v_{ij} + \delta_2 \ln v_{ij}^{revision} + \delta_3 \ln v_{ij}^{arthroscopy} - \tau \ln d_{ij} + \eta_{ij}$, where $v_{ij}^{revision}$ and $v_{ij}^{arthroscopy}$ denote surgeon j's hip and knee revision and knee arthroscopy volume plus one, respectively.

	(1) Complication	(2) Complication	(3) Complication	(4) Complication	(5) Complication
Log volume $(\ln v_{ij})$	-0.0055	-0.0037	-0.0027	-0.0027	-0.0034
	(0.0003)	(0.0008)	(0.0006)	(0.0006)	(0.0008)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$		-0.0040	-0.0046	-0.0046	-0.0046
		(0.0018)	(0.0018)	(0.0018)	(0.0018)
Roy selection $(\theta_{ij}(j))$				-0.0002	-0.0002
				(0.0004)	(0.0004)
N	696,620	696,620	696,620	696,620	696,620
Mean complication	0.031	0.031	0.031	0.031	0.031
Standard deviation surgeon selection term $(\widehat{\phi}_k)$			0.014	0.014	0.014
Risk covariates	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓
Surgeon HRR fixed effects	X	✓	✓	✓	✓
Measurement error correction	X	X	X	X	✓

Table D3: Progressively adding selection correction terms and measurement error correction

Notes: This table shows the estimates from estimating Equation 7. The outcome is a binary indicator equal to one if a patient has a complication. All specifications are estimated using a linear probability model. Column (2) adds a control for the estimated demand for a surgeon's exogenous quality, $\hat{\delta}_{0j}$, as well as fixed effects for a surgeon's HRR. Column (3) adds the surgeon-specific selection correction terms, θ_{ik}'s. Column (4) adds the Roy selection term, θ_{ij} Column (5) adjusts for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5 for columns (2)-(5).

	(1) Complication	(2) Complication	(3) Complication	(4) Complication	(5) Complication
Log surgeon volume $(\ln v_{ij})$	-0.0020	-0.0020	-0.0018	-0.0016	-0.0016
	(0.0021)	(0.0021)	(0.0021)	(0.0021)	(0.0021)
Log hospital volume (excl. focal surgeon)		-0.0002	-0.0000	-0.0001	-0.0001
		(0.0008)	(0.0008)	(0.0008)	(0.0008)
Log anesthesiologist volume			-0.0007	-0.0004	-0.0003
			(0.0007)	(0.0008)	(0.0008)
Surgeon-anesthesiologist prior experience				-0.0011	-0.0011
				(0.0011)	(0.0011)
Cemented replacement					-0.0066
					(0.0032)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$	-0.0033	-0.0032	-0.0032	-0.0032	-0.0032
	(0.0046)	(0.0047)	(0.0047)	(0.0047)	(0.0047)
Roy selection $(\theta_{ij}(j))$	0.0012	0.0013	0.0012	0.0012	0.0012
	(0.0016)	(0.0017)	(0.0017)	(0.0017)	(0.0017)
N	121,765	121,765	121,765	121,765	121,765
Mean complication	0.027	0.027	0.027	0.027	0.027
Standard deviation surgeon selection term $(\widehat{\phi}_k)$	0.250	0.249	0.247	0.245	0.237
Risk covariates	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓
Surgeon HRR fixed effects	✓	✓	✓	✓	✓
Measurement error correction	✓	✓	✓	✓	✓

Table D4: Further examining specific mechanisms underlying the returns to surgeon volume

Notes: This table shows the estimates from estimating Equation 7. The outcome is a binary indicator equal to one if a patient has a complication. All specifications are estimated using a linear probability model. Column (2) adds a control for the hospital's log volume (less the surgeon's volume). Column (3) adds a control for the anesthesiologist's log volume. Column (4) adds a binary control equal to one if the surgeon and anesthesiologist have worked together in the previous 365 days. Column (5) adds a control for if the surgeon uses a cemented approach. The coefficients on log volume are adjusted for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5.

	(1)	(2)
Model	Control function	Control function
Outcome	Complication	Complication
Log volume $(\ln v_{ij})$	-0.0030	-0.0030
	(0.0011)	(0.0011)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$	-0.0058	-0.0056
	(0.0027)	(0.0028)
Roy selection $(\theta_{ij}(j))$	-0.0003	-0.0002
	(0.0010)	(0.0010)
N	487,238	487,238
Mean complication	0.031	0.031
Standard deviation surgeon selection term $(\widehat{\phi}_k)$	0.134	0.136
Risk covariates	✓	✓
Year fixed effects	✓	✓
Surgeon HRR fixed effects	✓	✓
Measurement error correction	✓	✓
Surgeon covariates	X	✓
Market covariates	X	✓

Table D5: Estimate of the returns to surgeon volume with surgeon and market covariates

Notes: This table shows the estimates from estimating Equation 7. The outcome is a binary indicator equal to one if a patient has a complication. Both specifications are estimated using a linear probability model. The specification in column (1) omits surgeon and market-specific controls. Column (2) adds these controls, which include three dummies for the rank of a surgeon's medical school (1–25 rank; 26–50 rank; 51+ rank), log HSA market size, log ZIP code market size, the log distance between the surgeons' medical school and their practice locations, and the ZIP code amenity index as described in Section C.6. The coefficients on log volume are adjusted for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5.

	(1)	(2)
Model	Control function	Control function
Outcome	Complication	Complication
Log volume $(\ln v_{ij})$	-0.0033	-0.0032
	(0.0008)	(0.0008)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$	-0.0047	-0.0050
	(0.0018)	(0.0023)
Roy selection $(\theta_{ij}(j))$	-0.0001	-0.0011
	(0.0004)	(0.0016)
N	692,446	692,446
Mean complication	0.031	0.031
Standard deviation surgeon selection term $(\widehat{\phi}_k)$	0.015	0.015
Risk covariates	✓	✓
Year fixed effects	✓	✓
Surgeon HRR fixed effects	✓	✓
Measurement error correction	✓	✓
Patient ZIP code fixed effects	X	✓

Table D6: Estimate of the returns to surgeon volume with fixed effects for patient ZIP code of residence

Notes: This table shows the estimates from estimating Equation 7. The outcome is a binary indicator equal to one if a patient has a complication. All specifications are estimated using a linear probability model. Column (2) adds fixed effects for a patient's ZIP code of residence. The coefficients on log volume are adjusted for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5.

Model	(1) Control function	(2) Control function	(3) Control function
Outcome	Complication	Complication	Complication
Log volume $\left(\ln v_{ij}\right)$	-0.0032	-0.0025	-0.0023
	(0.0008)	(0.0008)	(0.0007)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$	-0.0040	-0.0055	
	(0.0019)	(0.0031)	
Roy selection $\left(\theta_{ij}(j)\right)$	0.0000	-0.0003	-0.0011
	(0.0004)	(0.0007)	(0.0007)
N	654,743	654,743	654,743
Mean complication	0.031	0.031	0.031
Standard deviation surgeon selection term $\left(\widehat{\phi}_k\right)$	0.019	0.023	0.075
Risk covariates	✓	✓	✓
Year fixed effects	✓	✓	✓
Surgeon HRR fixed effects	✓	✓	✓
Measurement error correction	✓	✓	✓
Hospital fixed effects	X	✓	X
Surgeon fixed effects	X	X	✓

Table D7: Estimates of the returns to surgeon volume with hospital and surgeon fixed effects

Notes: This table shows the estimates from estimating Equation 7. The outcome is a binary indicator equal to one if a patient has a complication. All specifications are estimated using a linear probability model. Column (2) adds hospital fixed effects, and column (3) adds surgeon fixed effects. The coefficients on log volume are adjusted for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5.

	(1)	(2)
Model	Control function	Control function
Outcome	Complication	Complication
Log volume $(\ln v_{ij})$	-0.0034	-0.0031
	(0.0008)	(0.0008)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$	-0.0046	-0.0048
	(0.0018)	(0.0019)
Roy selection $(\theta_{ij}(j))$	-0.0002	-0.0002
	(0.0004)	(0.0004)
N	696,620	683,148
Mean complication	0.031	0.031
Standard deviation surgeon selection term $(\widehat{\phi}_k)$	0.014	0.017
Risk covariates	✓	✓
Year fixed effects	✓	✓
Surgeon HRR fixed effects	✓	✓
Measurement error correction	✓	✓

Table D8: Robustness to adding one to hip and knee volume

Notes: This table shows the estimates from estimating Equation 7. The outcome is a binary indicator equal to one if a patient has a complication. All specifications are estimated using a linear probability model. Column (2) replaces log (plus one) hip and knee volume with log hip and knee volume. The coefficients on log volume are adjusted for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5.

	(1)	(2)
Model	OLS	Control function
Outcome	Complication	Complication
2nd volume quintile [30–45]	-0.0040	-0.0019
	(0.0007)	(0.0010)
3rd volume quintile [50–80]	-0.0066	-0.0026
	(0.0007)	(0.0014)
4th volume quintile [85–135]	-0.0097	-0.0052
	(0.0007)	(0.0017)
5th volume quintile [140–845]	-0.0142	-0.0082
	(0.0007)	(0.0021)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$		-0.0048
		(0.0019)
Roy selection $\left(\theta_{ij}(j)\right)$		-0.0001
		(0.0004)
N	696,620	696,620
Mean complication	0.031	0.031
Standard deviation surgeon selection term $\left(\widehat{\phi}_k\right)$		0.014
Risk covariates	✓	✓
Year fixed effects	✓	✓
Surgeon HRR fixed effects	X	✓
Measurement error correction	X	✓

Table D9: Robustness to using volume quintiles

Notes: This table shows the estimates from estimating Equation 7 except volume quintiles are used instead of log hip and knee volume. The reference group is the first volume quintile [0–30]. The outcome is a binary indicator equal to one if a patient has a complication. All specifications are estimated using a linear probability model. The specification in column (1) is the OLS specifications, whereas column (2) instruments for volume using differential distance using the control function approach. The coefficients on volume are adjusted for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5 for column (2).

	(1)	(2)
Model	Control function	Control function
Outcome	Complication	Complication
Log volume $(\ln v_{ij})$	-0.0034	
	(0.0008)	
Log knee volume		-0.0020
		(0.0007)
Log hip volume		-0.0017
		(0.0007)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$	-0.0046	-0.0046
	(0.0018)	(0.0019)
Roy selection $(\theta_{ij}(j))$	-0.0002	-0.0002
	(0.0004)	(0.0004)
N	696,620	696,620
Mean knee complication	0.025	0.025
Mean hip complication	0.044	0.044
Standard deviation surgeon selection term $(\widehat{\phi}_k)$	0.014	0.014
Risk covariates	✓	✓
Year fixed effects	✓	✓
Surgeon HRR fixed effects	✓	✓
Measurement error correction	✓	✓

Table D10: Separating out hip and knee volume

Notes: This table shows the estimates from estimating Equation 7. The outcome is a binary indicator equal to one if a patient has a complication. All specifications are estimated using a linear probability model. Column (2) replaces log hip and knee volume with log hip volume and log knee volume separately. The coefficients on log volume are adjusted for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5.

	(1)	(2)
Model	Control function	Control function
Outcome	Complication	Complication
Log volume $(\ln v_{ij})$	-0.0034	-0.0031
	(0.0008)	(0.0008)
Log revision volume		-0.0007
		(0.0005)
Log knee arthroscopy volume		-0.0000
		(0.0004)
Demand for exogenous quality $\left(\widehat{\delta}_{0j}\right)$	-0.0046	-0.0044
	(0.0018)	(0.0019)
Roy selection $(\theta_{ij}(j))$	-0.0002	-0.0001
	(0.0004)	(0.0004)
N	696,620	696,620
Mean complication	0.031	0.031
Standard deviation surgeon selection term $(\widehat{\phi}_k)$	0.014	0.014
Risk covariates	✓	✓
Year fixed effects	✓	✓
Surgeon HRR fixed effects	✓	✓
Measurement error correction	✓	✓

Table D11: Controlling for hip and knee revisions and knee arthroscopies

Notes: This table shows the estimates from estimating Equation 7. The outcome is a binary indicator equal to one if a patient has a complication. All specifications are estimated using a linear probability model. Column (2) adds controls for a surgeon's log hip and knee arthroplasty revision volume, as well as their knee arthroscopy volume. The coefficients on log volume are adjusted for measurement error due to random sampling of the Medicare FFS claims, as discussed in Appendix C.4. Standard errors are clustered at the surgeon level and are calculated as described in Appendix C.5.

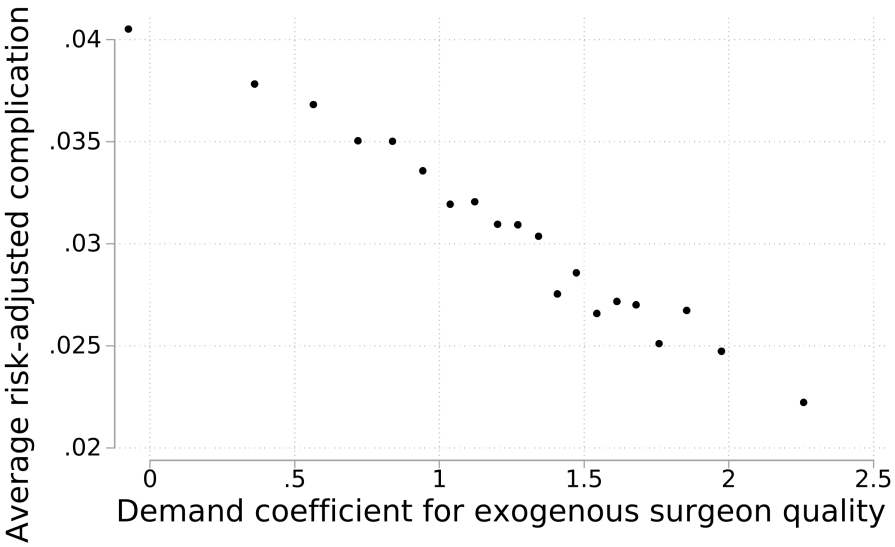


Figure D1: Relationship between risk-adjusted complication rate and demand for exogenous surgeon quality

Notes: This figure shows the relationship between the risk-adjusted complication rate and the demand for a surgeon's exogenous quality, $\widehat{\delta}_{0j}$, as estimated in Equation 1. The unit of observation is a surgeon-HRR. The risk-adjusted complication rate is shrunk to the national mean using the Empirical Bayes procedure described in Appendix Section C.1. Meanwhile, δ_{0j} is shrunk to the market-level mean using the same procedure. HRR fixed effects have been residualized out.

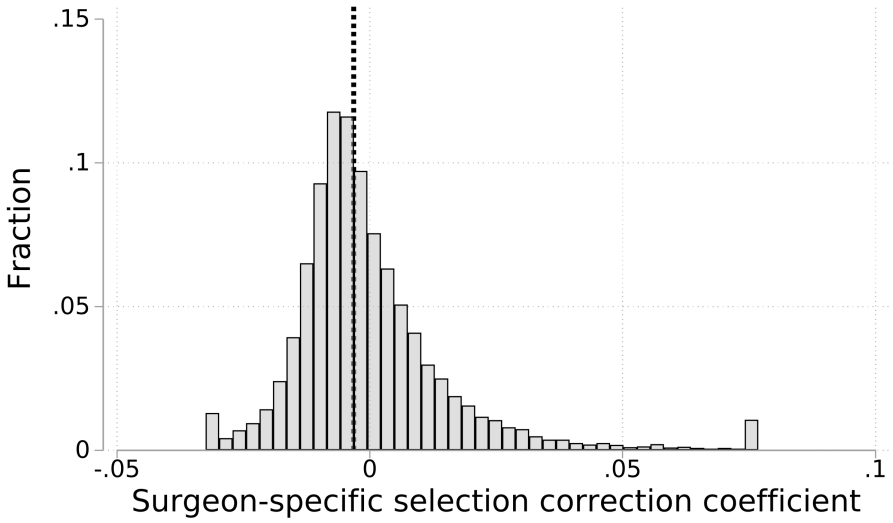


Figure D2: Distribution of surgeon-specific selection correction coefficients ($\widehat{\phi}_k$)

Notes: This figure shows the distribution of surgeon-specific selection correction terms, $\widehat{\phi}_k$, as estimated from Equation 7. These surgeon-specific selection correction terms have been winsorized at the 1% level. The vertical line denotes the median surgeon-specific selection correction term.

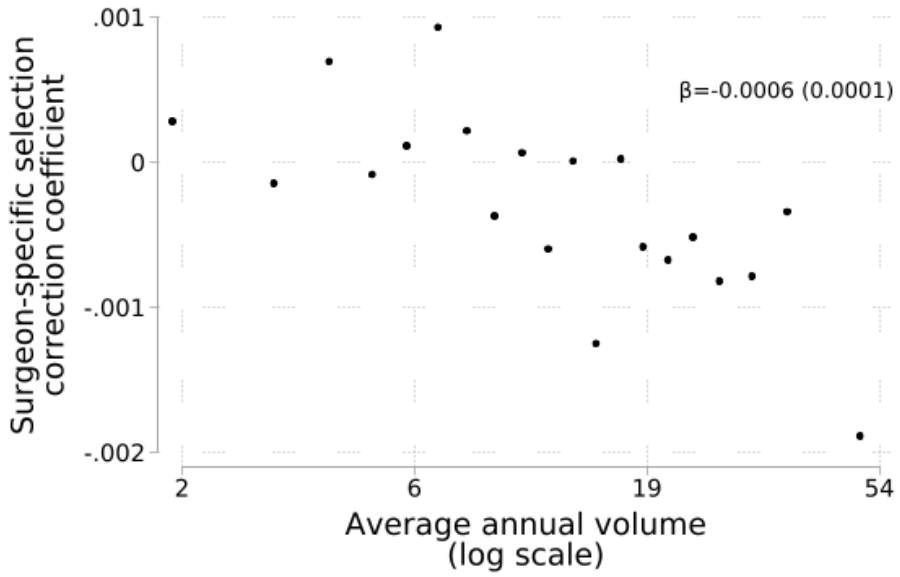


Figure D3: Relationship between surgeon-specific selection correction term ($\widehat{\phi}_k$) and average surgeon hip and knee volume

Notes: This figure shows the relationship between the average surgeon-specific selection correction term, $\widehat{\phi}_k$, as estimated from Equation 7, and a surgeon's average hip and knee volume. The surgeon-specific selection correction terms have been winsorized at the 1% level. The unit of observation is a surgeon-HRR, and each observation is weighted by cumulative hip and knee volume. HRR fixed effects have been residualized out.

E Policy implications appendix

E.1 Defining consumer surplus and converting miles to dollars

To expound on the welfare measure, I formally define consumer surplus. Consumer surplus is the dollar-denominated sum of patients' expected utilities from the surgeons in their choice sets, assuming all patients have equal welfare weights. Formally, for patient i and surgeon j in market m, consumer surplus is:

$$CS_m = \sum_{i \in m} I_i \times \frac{(1 + \overline{d_i})t_c}{\tau_m}, \tag{E1}$$

where $I_i = \ln\left(\sum_j \exp\left(\delta_j(v_{ij}) - \tau \ln d_{ij}\right)\right)$ is the inclusive value or expected utility, $\overline{d_i}$ is the probability-weighted sum of the pre-policy distances to each surgeon in patient i's choice set, t_c is the dollar value of distance constructed based on the average wage in patient i's county of residence c, and τ_m is the estimated distance coefficient in market m from Equation 1.[37] The second term, $\frac{(1+\overline{d_i})t_c}{\tau_m}$, acts as a conversion factor between utils and miles, where $\frac{1+\overline{d_i}}{\tau_m}$ is the marginal rate of substitution between utils and miles at patient i's expected travel distance and t_c is the dollar cost of a mile. I also use this conversion factor to dollar-denominate the first-best Pigouvian taxes and subsidies.[38] In expressing consumer surplus in this way, I assume that patients fully internalize the cost of a complication to themselves, as they trade off a surgeon's quality with the distance they must travel.

Following the literature, I calculate the dollar cost of a mile in county c, t_c, as the dollar value of driving plus the opportunity cost of time (Einav, Finkelstein, and Williams, 2016; Dolfen et al., 2023). Specifically, I compute the dollar value of driving using the Internal Revenue Service (IRS) mileage reimbursement rate and the opportunity cost of time using the average wage in a patient's county:

$$t_c = \underbrace{2}_{\text{round trip}} \times \left(\underbrace{\frac{10.9 \text{ minutes}}{5.3 \text{ miles}}}_{\substack{\text{minutes/} \\ \text{straight-line mile}}} \times \underbrace{\text{Wage}_c}_{\$/\text{minute}} + \underbrace{\frac{7.9 \text{ drive miles}}{5.3 \text{ miles}}}_{\substack{\text{drive miles/} \\ \text{straight-line mile}}} \times \underbrace{\frac{0.55 \text{ dollars}}{\text{drive miles}}}_{\substack{\text{IRS mileage} \\ \text{reimbursement rate}}} \right). \tag{E2}$$

Here, the number outside the parentheses accounts for the round trip that patients must travel for their hip or knee replacement. Inside the parentheses is the dollar cost of a mile, where the first term before the sum is the lost wages from travel and the second is the cost of fuel and depreciation. I obtain the two conversion factors—between straight-line miles and minutes and between straight-line miles and drive miles—from Einav, Finkelstein, and Williams (2016). I also compute

[37] To limit the influence of noisier estimates of τ in smaller markets, I shrink them to the national mean using the Empirical Bayes procedure described in Appendix C.1.

[38] In this case, though, I just use the average distance to each surgeon for all patients in the sample for $\overline{d_i}$.

the fiscal costs of transportation subsidies as t_c multiplied by the patient's expected travel distance under the subsidy and the subsidy rate. This formulation imposes the assumption that the opportunity cost of time is homogeneous across patients within a county and that the opportunity cost of time for a Medicare hip and knee patient is the same as that of a working individual.

E.2 Simplified first-best policy: Two surgeons at the ends of a Hotelling line

To provide intuition for the calculation of the first-best policy, I calculate the first-best policy in a stylized example. Consider two surgeons, surgeon A and surgeon B, with the same exogenous quality on the two endpoints of a Hotelling line, as depicted in Figure E1. Because it is a Hotelling line, patients are uniformly distributed between the two surgeons. The social planner's objective is to assign patients to surgeons to maximize welfare. In this case, this objective corresponds to choosing the share of patients who surgeon A operates on, as this share pins down exactly what share surgeon B operates on, since there are only two surgeons and a uniform mass of patients. To keep this example illustrative, I define welfare in this stylized example as consumer surplus, and I remove the logs from the demand model such that $u_{ij} = \delta_{0j} + \delta_1 v_{ij} - \tau d_{ij} + \eta_{ij}$. I also assume surgeons have no capacity constraints. These assumptions substantially simplify the calculations but still convey the primary message.

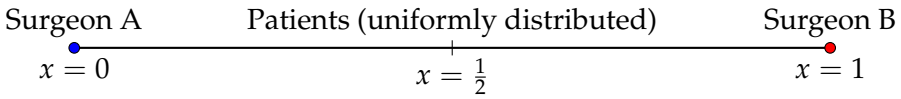


Figure E1: Patients uniformly distributed between two surgeons on a Hotelling line

I define this problem formally so as to solve for the solution mathematically and build intuition. Let u_i^A denote the utility of patient i from surgeon A and u_i^B denote the utility of that same patient from surgeon B. Since the mean of the η's is a constant, the social planner's objective is:

$$\begin{aligned}\max_{s\in[0,1]} W(s) &= \int_0^s u_i^A + \int_s^1 u_i^B \\ &= \int_0^s (\delta_0 + \delta_1 s - \tau x)dx + \int_s^1 (\delta_0 + \delta_1(1-s) - \tau(1-x))dx, \end{aligned} \tag{E3}$$

where s is the share of patients assigned to surgeon A and x is the distance between a patient and surgeon A. These definitions imply that the share of patients assigned to surgeon B is $1 - s$, and the distance from a patient to surgeon B is $1 - x$.

I now solve for the optimal allocation of patients to surgeons in this stylized example under two separate conditions. First, consider the example where there are no returns to surgeon volume, such that $\delta_1 = 0$. Then $s^* = \frac{1}{2}$, since the surgeons are identical, so the social planner should just assign patients to the closest surgeon. Now, suppose that $\delta_1 > 0$. In this case, the solution is less obvious, as there is now a tradeoff between sending a patient farther and boosting (or lowering) a surgeon's volume, which increases (or decreases) everyone else's utility who is assigned to

that surgeon. To proceed, I re-express the equation for welfare above:

$$\begin{aligned} W(s) &= \int_0^s (\delta_0 + \delta_1 s - \tau x)dx + \int_s^1 (\delta_0 + \delta_1(1-s) - \tau(1-x))dx \\ &= s\left[\delta_0 + \delta_1 s - \frac{\tau}{2}s\right] + (1-s)(\delta_0 + \delta_1(1-s)) - \frac{\tau}{2}(1-s)^2 \\ &= \delta_0 + \left(\delta_1 - \frac{\tau}{2}\right)(2s^2 - 2s + 1) \end{aligned}$$

Solving for the first-order condition (FOC) then implies:

$$\left(\delta_1 - \frac{\tau}{2}\right)(4s - 2) = 0$$

Thus, solving for the optimal s^* yields:

$$s^* = \begin{cases} \text{any } s \in [0,1], & \text{if } \delta_1 = \frac{\tau}{2} \\ \frac{1}{2}, & \text{if } \delta_1 < \frac{\tau}{2} \\ 0 \text{ or } 1, & \text{if } \delta_1 > \frac{\tau}{2} \end{cases}. \tag{E4}$$

I arrive at the first condition since any $s \in [0,1]$ satisfies the FOC if $\delta_1 = \frac{\tau}{2}$. I arrive at the next two, assuming $4s - 2 = 0$. s^* is a local maximum if $\delta_1 - \frac{\tau}{2} < 0$. However, it is a local minimum if $\delta_1 - \frac{\tau}{2} > 0$, implying that a border solution with $s = 0$ or $s = 1$ solves the maximization problem.

This solution lends insight into the conditions that determine how the social planner optimally assigns patients. The optimal assignment depends on the relative importance of the returns to surgeon volume relationship, as captured by δ_1, and the cost of travel, as captured by τ. When the returns to surgeon volume relationship perfectly offsets the distance cost, any allocation of patients is optimal. When the returns to surgeon volume relationship is small relative to the distance cost, the planner should just assign patients to the closest surgeon. In this case, assigning a farther away patient to a surgeon does not generate sufficient public benefit to justify the larger travel cost. Finally, when the returns to surgeon volume relationship is large relative to the distance cost, allocating all patients to one surgeon outweighs the costs of having patients travel farther.

E.3 Computing compensated changes in consumer surplus

For the policy counterfactual in which I subsidize patient transportation, I compute the welfare change using a compensated change in consumer surplus to isolate welfare changes resulting from the reallocation of patients to surgeons. Under this subsidy, patients may have higher utility for two reasons: they are now effectively closer to all surgeons (an "income effect") and they may choose different surgeons (a "substitution effect").[39] In this counterfactual analysis, I compute only the substitution effect, since it captures the welfare change only from the reallocation of patients to surgeons and thus how the policy may address the externality of patient choice. To isolate this effect, I hold fixed the baseline utility, such that the consumer surplus from the choice set with the subsidy is the same as that without. Formally, with some vector of surgeon volumes $\vec{v}$, I compute the change in consumer surplus, CS, under some subsidy s in market m as:

$$\begin{aligned}\Delta CS_m(s,\vec{v}) &= \underbrace{[CS_m(s=s,\vec{v}=\vec{v}_s) - CS_m(s=0,\vec{v}=\vec{v}_0)]}_{\text{full change in utility}} - \underbrace{[CS_m(s=s,\vec{v}=\vec{v}_0) - CS_m(s=0,\vec{v}=\vec{v}_0)]}_{\text{"income effect": patients closer to all surgeons}} \\ &= \underbrace{CS_m(s=s,\vec{v}=\vec{v}_s) - CS_m(s=s,\vec{v}=\vec{v}_0)}_{\text{"substitution effect": patients reallocate}} \end{aligned} \tag{E5}$$

where $\vec{v}_0$ is the vector of surgeon hip and knee volumes when $s = 0$ (i.e., there is no subsidy) and $\vec{v}_s$ is the vector of surgeon hip and knee volumes when $s = s$. Intuitively, therefore, I compensate patients for the mechanical increase in utility from surgeons being effectively closer, had they not been allowed to change their behavior.

[39] The income effect is non-zero because distance enters the utility function non-linearly, but the government values distance linearly. Computing the substitution effect therefore nets out this income effect.

E.4 Moving surgeons implementation details

I implement the policy that moves surgeons to shortage areas by moving surgeons whose primary practice location falls outside a shortage area into hospitals located in shortage areas within the same HRR. Specifically, I first identify all ZIP codes in my sample in which a hip or knee replacement is performed during my sample and is located in a geographic primary care Health Professional Shortage Area (HPSA) county.[40] I define eligible surgeon movers as those whose primary practice location falls outside of one of these HPSA ZIP codes.[41] I then assign one eligible surgeon to each HPSA ZIP code within the same HRR randomly, such that surgeons can only be moved within the same HRR.[42] Moving surgeons within the same HRR perhaps more accurately reflects how such a policy would actually affect the spatial distribution of surgeons, as surgeons may not move cross-county but rather within the same broader health care market. Additionally, this implementation facilitates the analysis, as the demand parameters δ_{0j} are only interpretable within an HRR.

[40] I focus on primary care HPSAs because there are no shortage designations for specialty care. The Health Resources and Services Administration (HRSA) designates primary care physician (PCP) shortage areas based on four statistics: the population-to-PCP ratio, the percent of the population below the federal poverty level, the infant health index (based on the infant mortality rate or the low birth weight rate), and the travel time to the nearest source of care outside the HPSA designation area. The geographic area ranges in size and can be a county, a group of counties, or part of a city.

[41] As before, I define surgeons' primary practice location as the ZIP code where they perform the plurality of their hip and knee replacements. This time, though, I define the primary practice location over the entire sample period.

[42] For HRRs where there are more destinations than movers, some destinations do not receive a mover. For HRRs with more movers than destinations, I assign surgeons randomly.

E.5 Tables

	No returns to surgeon volume externality	Returns to surgeon volume externality
$\Delta\mathcal{W}$ / person ($)	2065.14	1815.19
Δ (CS - G) / person ($)	-1.11	-17.43
Δ fiscal savings / person ($)	2066.24	1832.61

Table E1: Average welfare effect of optimal policy

Notes: This table shows the average welfare effect of implementing the optimal policy. The first row shows the average change in per-person welfare in dollars, as calculated using Equation 9. The second and third rows show the average change in the (per-person) sub-components of welfare: consumer surplus less fiscal spending on policy implementation and fiscal savings from reduced complications, respectively. Consumer surplus is defined in Equation E1. The fiscal costs of the policy are calculated as in Section 5.3 and are equal to zero. The fiscal savings per person are calculated as in Equation 9. The first column shows the effect without the returns to surgeon volume externality (i.e., setting $\delta_1 = 0$ from Equation 1 and $\beta_1 = 0$ from Equation 7), while the second column incorporates the returns to surgeon volume externality.

E.6 Figures

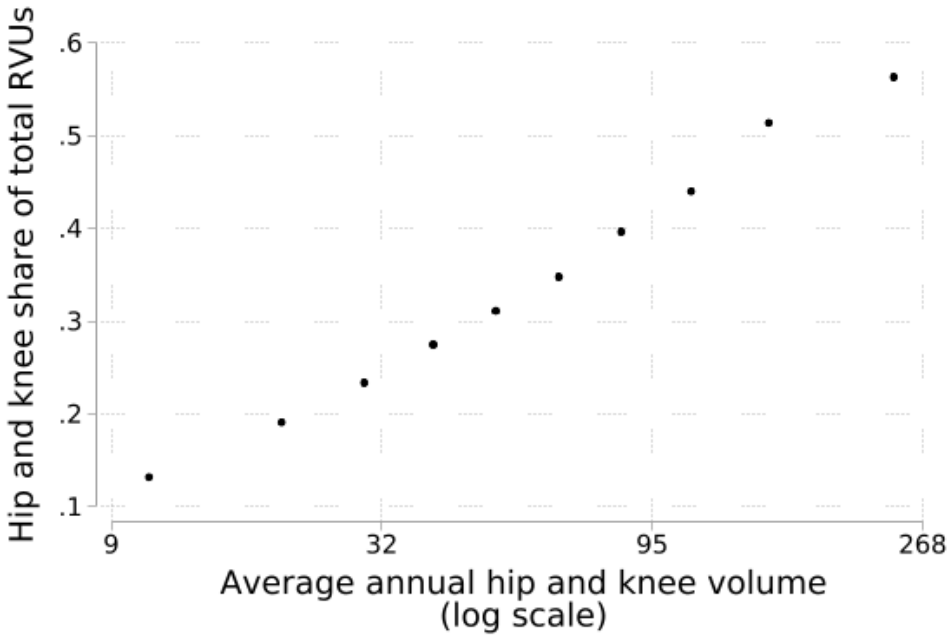


Figure E2: Relationship between share of hip and knee replacement RVUs and average annual hip and knee volume

Notes: This figure shows the relationship between the average annual share of a surgeon's relative value units (RVUs) allocated to hip and knee replacements, a proxy for work time spent on hip and knee replacements, and a surgeon's log average annual volume. The underlying unit of observation is a surgeon, which has been binned into ten equal-sized bins.

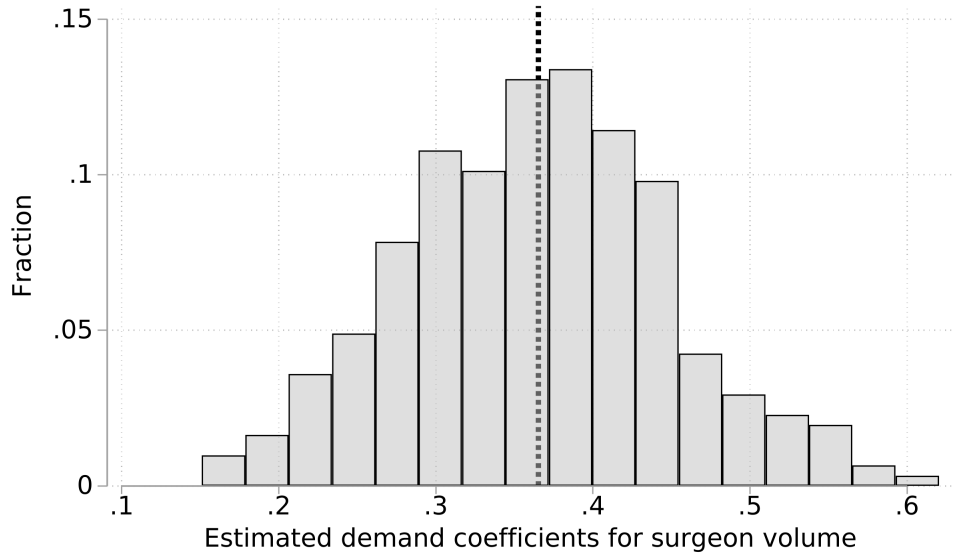


Figure E3: Histogram of estimated demand for surgeon volume across markets $\left(\widehat{\delta_1}\right)$

Notes: This figure shows the histogram of estimated volume coefficients for each HRR, as estimated from Equation 1. The coefficients have been shrunk to the national mean using the Empirical Bayes procedure described in Appendix Section C.1. The dotted vertical line denotes the mean.

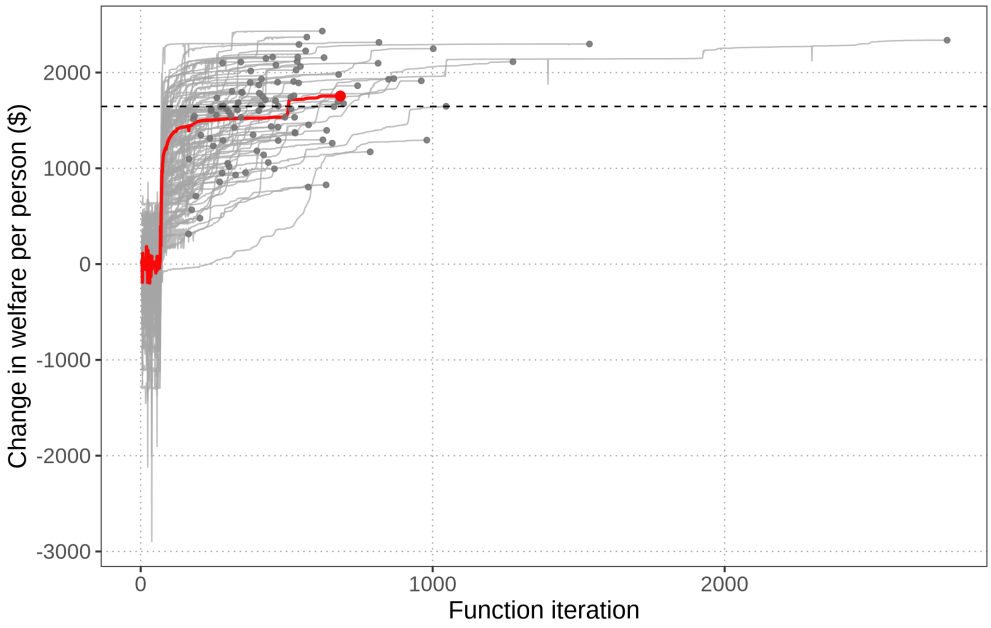


Figure E4: Welfare convergence paths for optimal policy under different start points without returns to surgeon volume externality in Springfield, Massachusetts

Notes: This figure shows the welfare convergence paths for 100 different starting tax vectors in one HRR (Springfield, Massachusetts). Each path denotes the change in welfare per person without the returns to surgeon volume externality under one of these starting vectors. The dots at the right end of each path indicate that the system has numerically converged. The red path corresponds to a start at the zero vector, which is the initialization used in the main analysis. The gray paths correspond to starting vectors drawn randomly from a standard normal distribution. The dashed horizontal line denotes the median final change in welfare across starts.

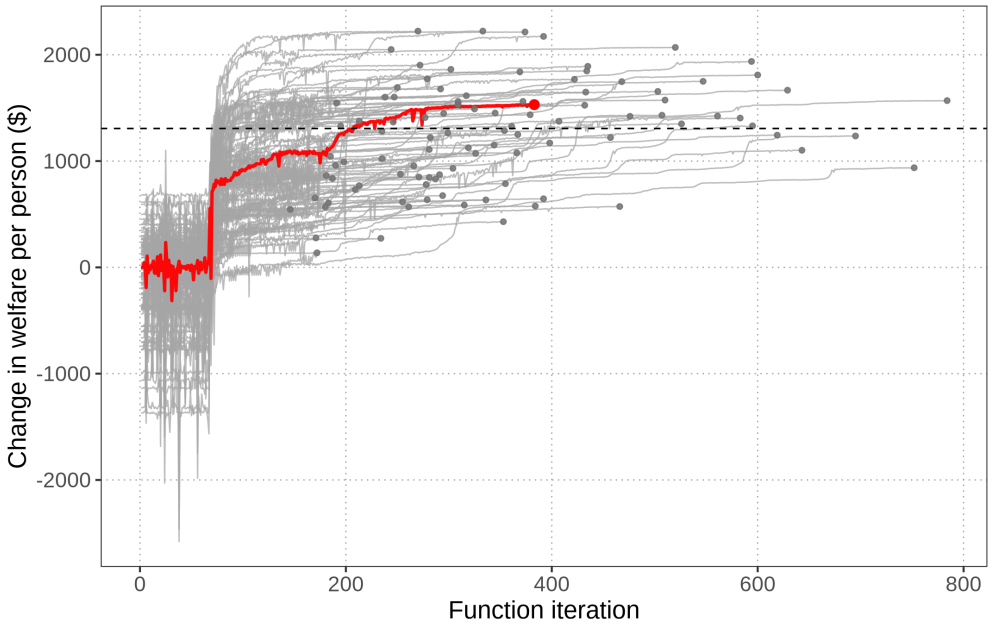


Figure E5: Welfare convergence paths for optimal policy under different start points with returns to surgeon volume externality in Springfield, Massachusetts

Notes: This figure shows the welfare convergence paths for 100 different starting tax vectors in one HRR (Springfield, Massachusetts). Each path denotes the change in welfare per person with the returns to surgeon volume externality under one of these starting vectors. The dots at the right end of each path indicate that the system has numerically converged. The red path corresponds to a start at the zero vector, which is the initialization used in the main analysis. The gray paths correspond to starting vectors drawn randomly from a standard normal distribution. The dashed horizontal line denotes the median final change in welfare across starts.

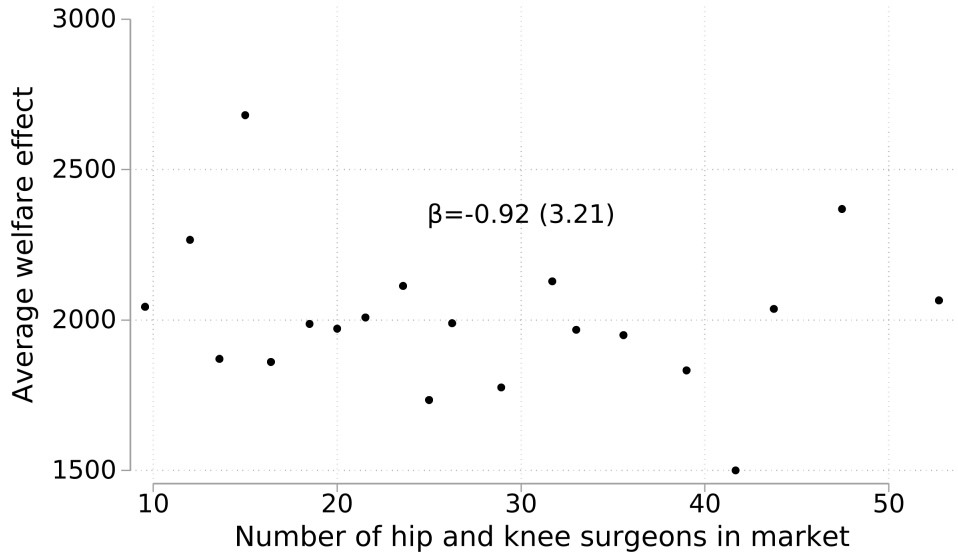


Figure E6: Relationship between welfare effect and number of hip and knee surgeons in market

Notes: This figure shows the relationship between the average welfare effect within a market and the number of hip and knee surgeons in that market. The welfare effect is the average of the effect with and without the returns to surgeon volume externality. The underlying unit of observation is an HRR.

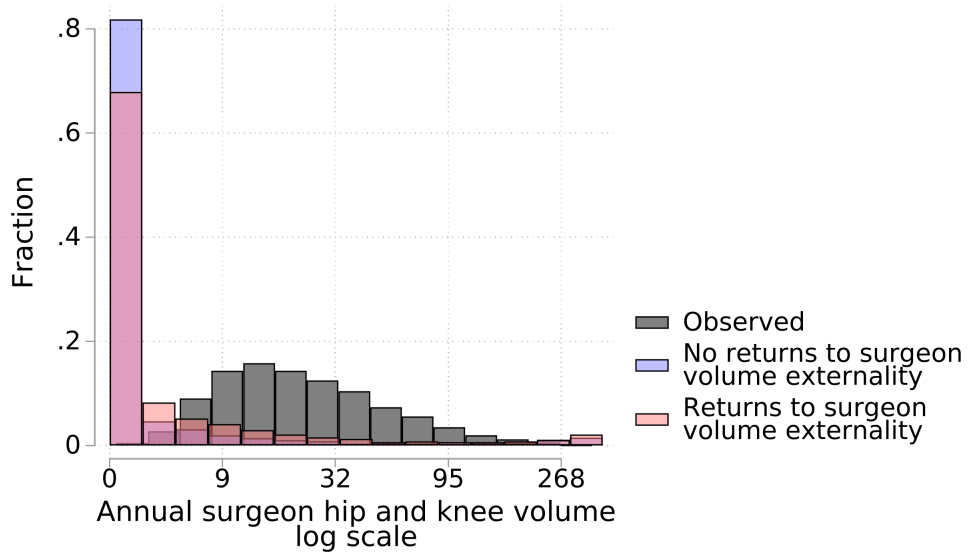


Figure E7: Histogram of observed and optimal annual hip and knee volume

Notes: This figure shows the histogram of observed volume, optimal volume without the returns to surgeon volume externality (i.e., setting $\delta_1 = 0$ from Equation 1 and $\beta_1 = 0$ from Equation 7), and optimal volume with the externality. The underlying unit of observation is a surgeon-HRR that has been binned into twenty-five equal-sized bins based on a surgeon's average log hip and knee volume after residualizing out HRR fixed effects. Observed volume above the capacity constraint has been trimmed from the histogram.

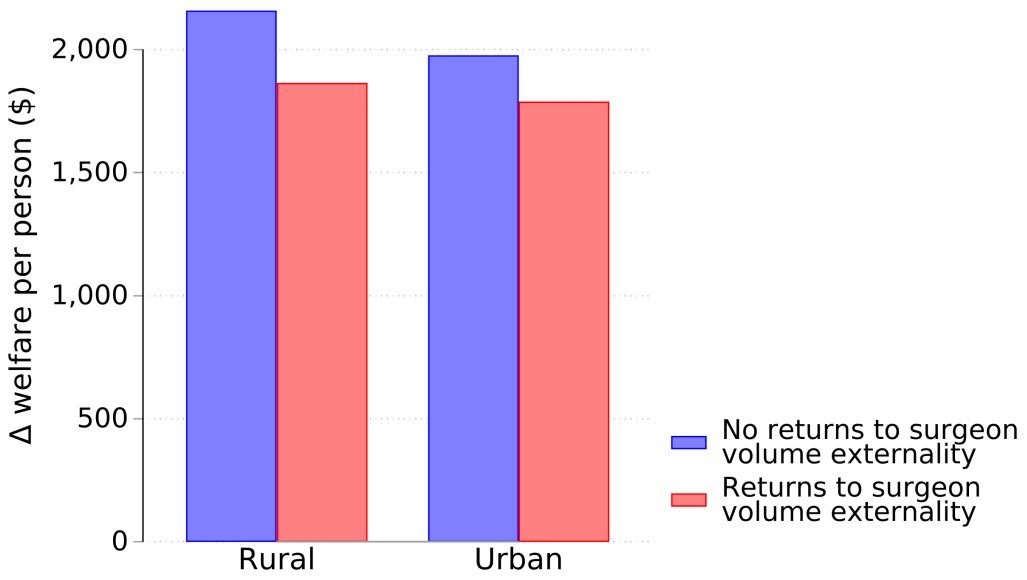


Figure E8: Welfare effects of optimal policy for rural versus urban patients

Notes: This figure shows the welfare effect of implementing the optimal policy for both rural and urban patients. Welfare is calculated in dollars using Equation 9. Rural patients are defined as those having ZIP codes of residence outside a "city center," where a "city center" is the HSA with the largest market size within an HRR. Urban patients are defined as those having ZIP codes of residence within a "city center." The blue bars show the effect without incorporating the returns to surgeon volume externality (i.e., setting $\delta_1 = 0$ from Equation 1 and $\beta_1 = 0$ from Equation 7), while the red bars incorporate this externality.

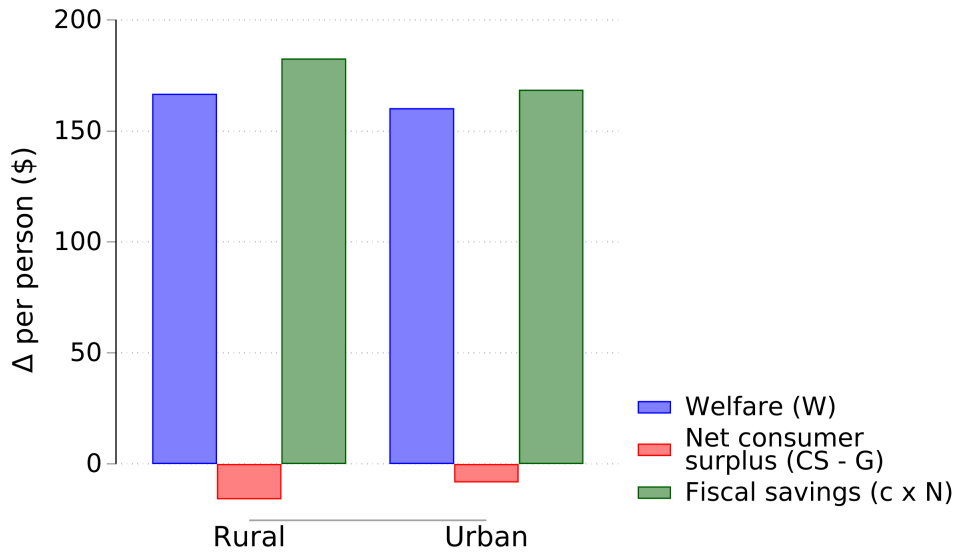


Figure E9: Effect of minimum volume standard on welfare, net consumer surplus, and fiscal savings for rural versus urban patients

Notes: This figure shows the effect of implementing a minimum volume standard on welfare, consumer surplus less the fiscal costs of policy implementation (net consumer surplus), and complication-related fiscal savings for both rural and urban patients. Welfare is calculated in dollars using Equation 9. Consumer surplus is defined in Equation E1. The fiscal costs of the policy are calculated as in Section 5.3 and are equal to zero for this counterfactual. The fiscal savings per person are calculated as in Equation 9. Rural patients are defined as those having ZIP codes of residence outside a "city center," where a "city center" is the HSA with the largest market size within an HRR. Urban patients are defined as those having ZIP codes of residence within a "city center." In 3 of the 306 HRRs, patients only live in one of the HSAs within the HRR, so these HRRs are omitted from this analysis. These effects incorporate the returns to surgeon volume externality.

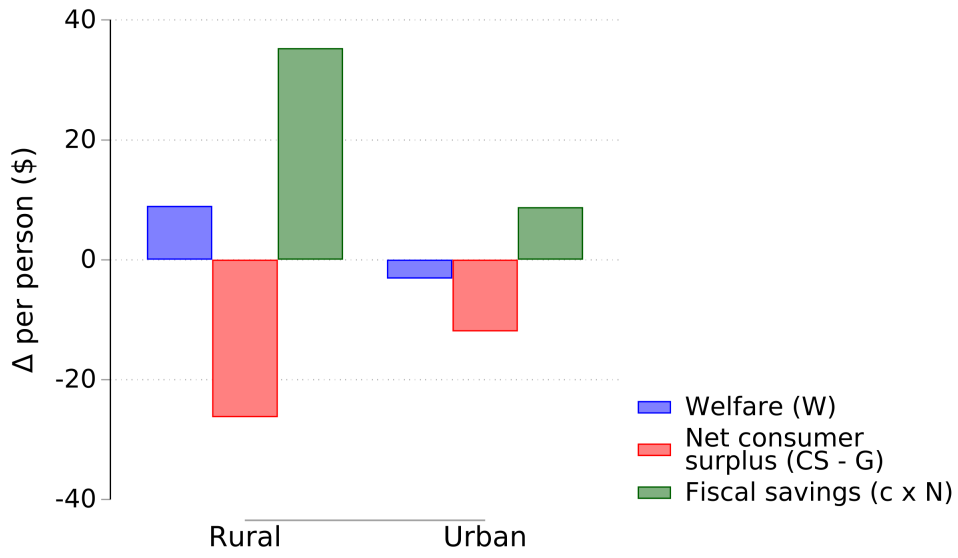


Figure E10: Effect of subsidizing transportation on welfare, net consumer surplus, and fiscal savings for rural versus urban patients

Notes: This figure shows the effect of subsidizing 30% of transportation costs on welfare, consumer surplus less the fiscal costs of policy implementation (net consumer surplus), and complication-related fiscal savings for both rural and urban patients. Welfare is calculated in dollars using Equation 9. Consumer surplus is defined in Equation E1. The fiscal costs of the policy are calculated as in Section 5.3. The fiscal savings per person are calculated as in Equation 9. Rural patients are defined as those having ZIP codes of residence outside a "city center," where a "city center" is the HSA with the largest market size within an HRR. Urban patients are defined as those having ZIP codes of residence within a "city center." In 3 of the 306 HRRs, patients only live in one of the HSAs within the HRR, so these HRRs are omitted from this analysis. These effects incorporate the returns to surgeon volume externality.

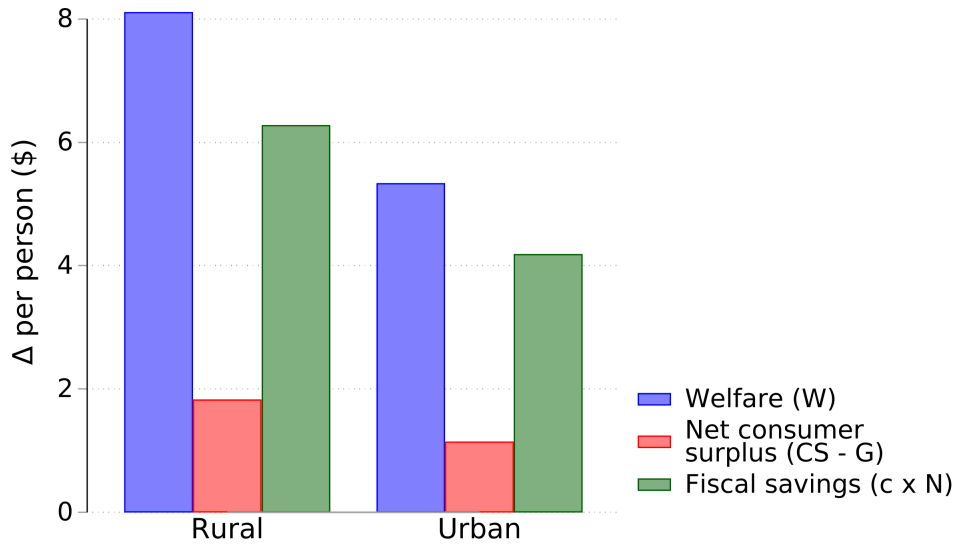


Figure E11: Effect of moving surgeons on welfare, net consumer surplus, and fiscal savings for rural versus urban patients

Notes: This figure shows the effect of moving surgeons to government-designated shortage areas on welfare, consumer surplus less the fiscal costs of policy implementation (net consumer surplus), and complication-related fiscal savings for both rural and urban patients. Welfare is calculated in dollars using Equation 9. Consumer surplus is defined in Equation E1. The fiscal costs of the policy are calculated as in Section 5.3 and are equal to zero for this counterfactual. The fiscal savings per person are calculated as in Equation 9. Rural patients are defined as those having ZIP codes of residence outside a "city center," where a "city center" is the HSA with the largest market size within an HRR. Urban patients are defined as those having ZIP codes of residence within a "city center." In 3 of the 306 HRRs, patients only live in one of the HSAs within the HRR, so these HRRs are omitted from this analysis. These effects incorporate the returns to surgeon volume externality.